

Class 4

Colin Kuehl

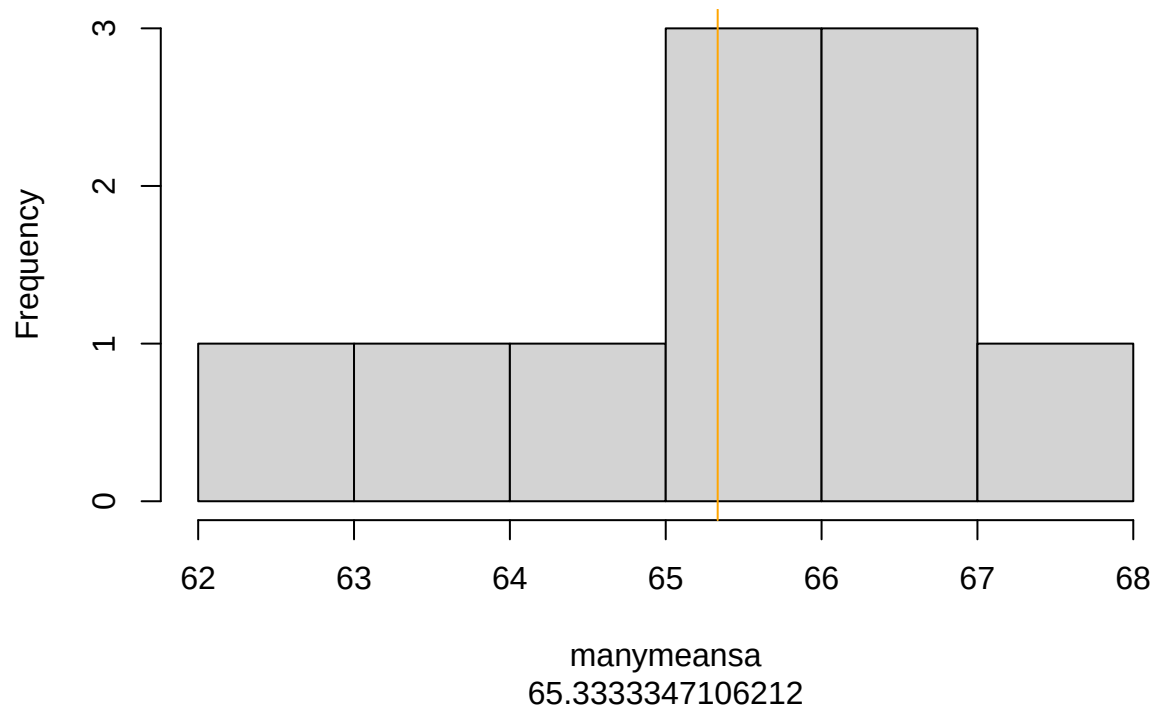
September 18, 2025

We'll start out illustrating concepts from discussion using R

Distribution of the Mean

```
getmean = function(N, mean, sd) {  
  opinion = rnorm(n = N, mean = mean, sd = sd)  
  meanopinion = mean(opinion)  
  return(meanopinion)  
}  
# Building a function in R. This is beyond the scope of the class  
  
manymeansa = replicate(n = 10, getmean(20, mean = 65, sd = 6))  
hist(manymeansa, main = "Many Means 10 Draws", sub = mean(manymeansa))  
mean(manymeansa)  
  
## [1] 65.33333  
  
abline(v = mean(manymeansa), col = "orange")
```

Many Means 10 Draws

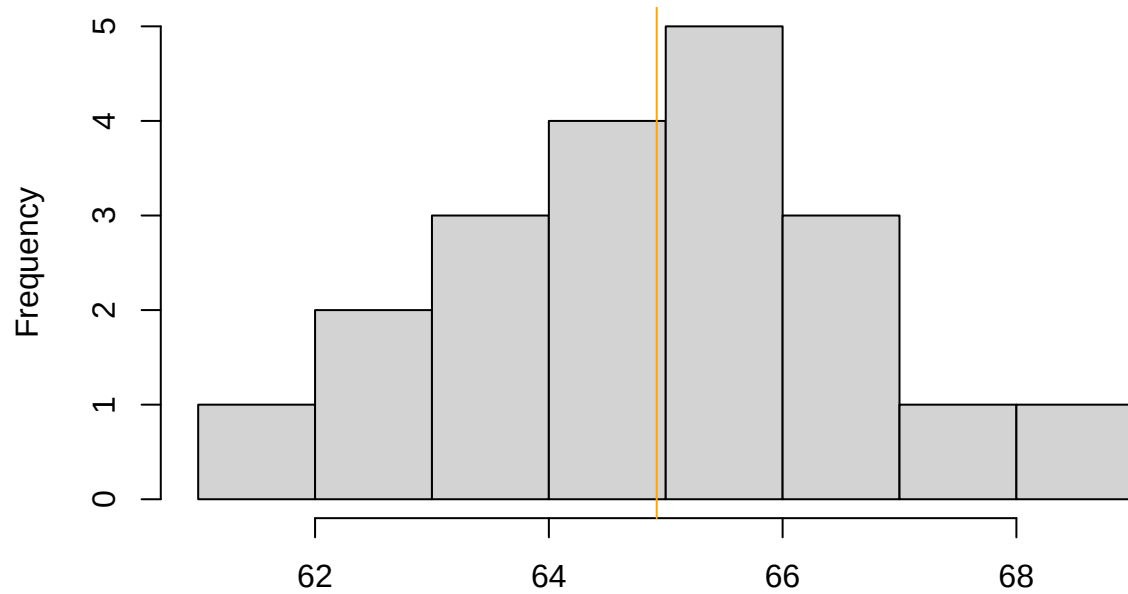


```
# look with different n
manymeansb = replicate(n = 20, getmean(20, mean = 65, sd = 6))
hist(manymeansb, main = "Many Means 20 Draws", sub = mean(manymeansb))
mean(manymeansb)
```

```
## [1] 64.92263
```

```
abline(v = mean(manymeansb), col = "orange")
```

Many Means 20 Draws



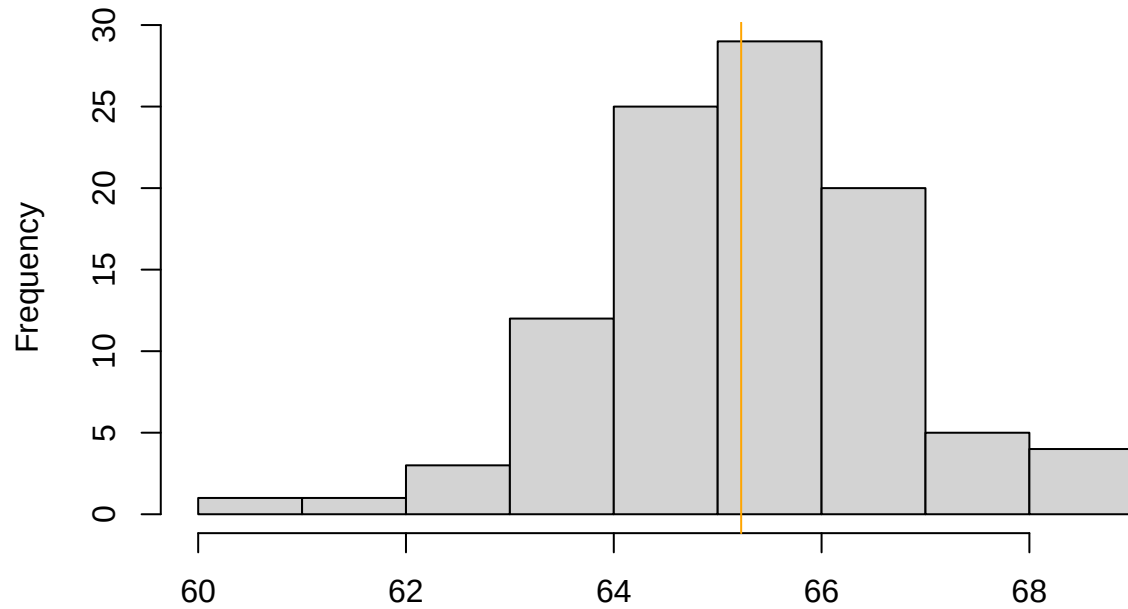
manymeansb
64.9226321673181

```
manymeansc = replicate(n = 100, getmean(20, mean = 65, sd = 6))  
hist(manymeansc, main = "Many Means 100 Draws", sub = mean(manymeansc))  
mean(manymeansc)
```

```
## [1] 65.22612
```

```
abline(v = mean(manymeansc), col = "orange")
```

Many Means 100 Draws



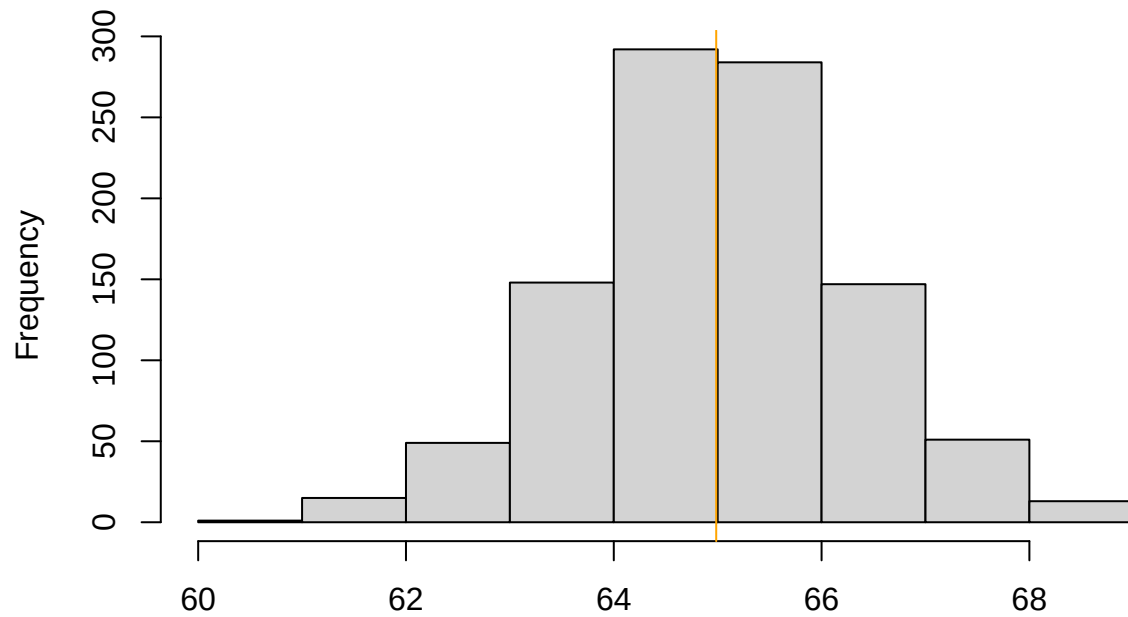
manymeansc
65.2261217458351

```
manymeansd = replicate(n = 1000, getmean(20, mean = 65, sd = 6))  
hist(manymeansd, main = "Many Means 1k Draws", sub = mean(manymeansd))  
mean(manymeansd)
```

```
## [1] 64.98577
```

```
abline(v = mean(manymeansd), col = "orange")
```

Many Means 1k Draws



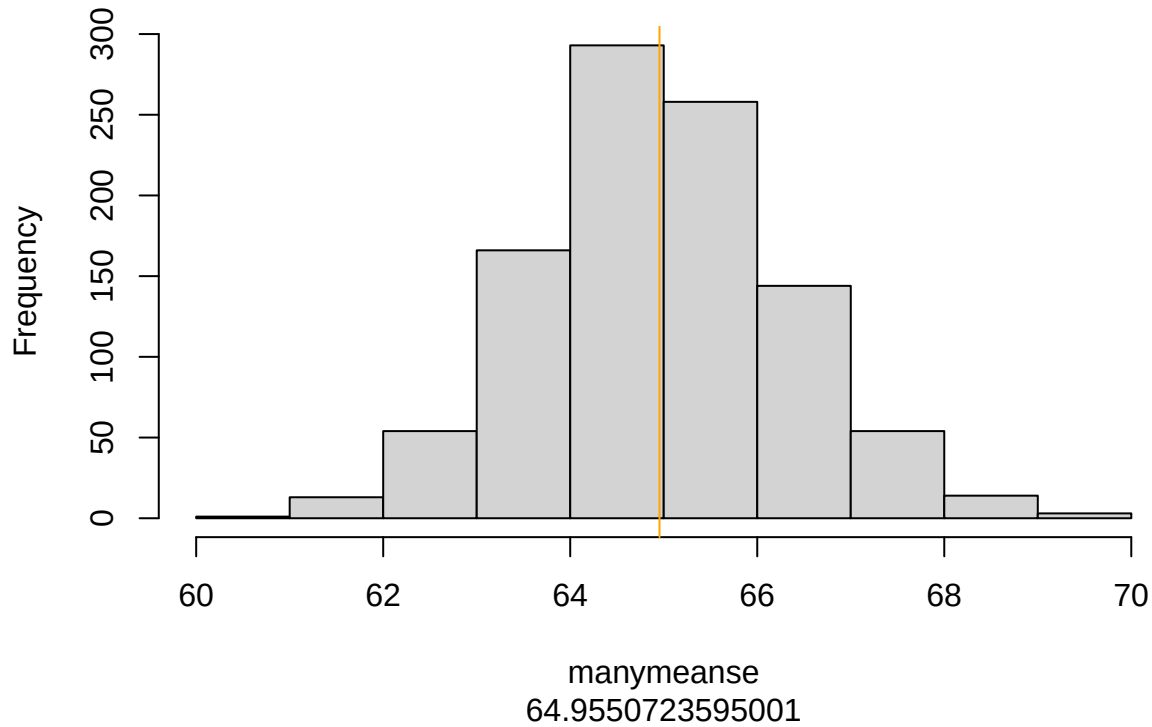
manymeansd
64.9857664805157

```
manymeanse = replicate(n = 1000, getmean(20, mean = 65, sd = 6))  
hist(manymeanse, main = "Many Means 1000 Draws", sub = mean(manymeanse))  
mean(manymeanse)
```

```
## [1] 64.95507
```

```
abline(v = mean(manymeanse), col = "orange")
```

Many Means 1000 Draws



```
manymeansf = replicate(n = 10000, getmean(20, mean = 65, sd = 6))
hist(manymeansf, main = "Many Means 10k Draws", sub = mean(manymeansf))
mean(manymeansf)
```

```
## [1] 64.99564
```

```
abline(v = mean(manymeansf), col = "orange")
```

```
# Larger sample sizes should be closer to 'true mean' but still variation
```

```
abline(v = 65, col = "blue")
```

```
abline(v = mean(manymeansa), col = "darkorange4")
```

```
abline(v = mean(manymeansb), col = "darkorange3")
```

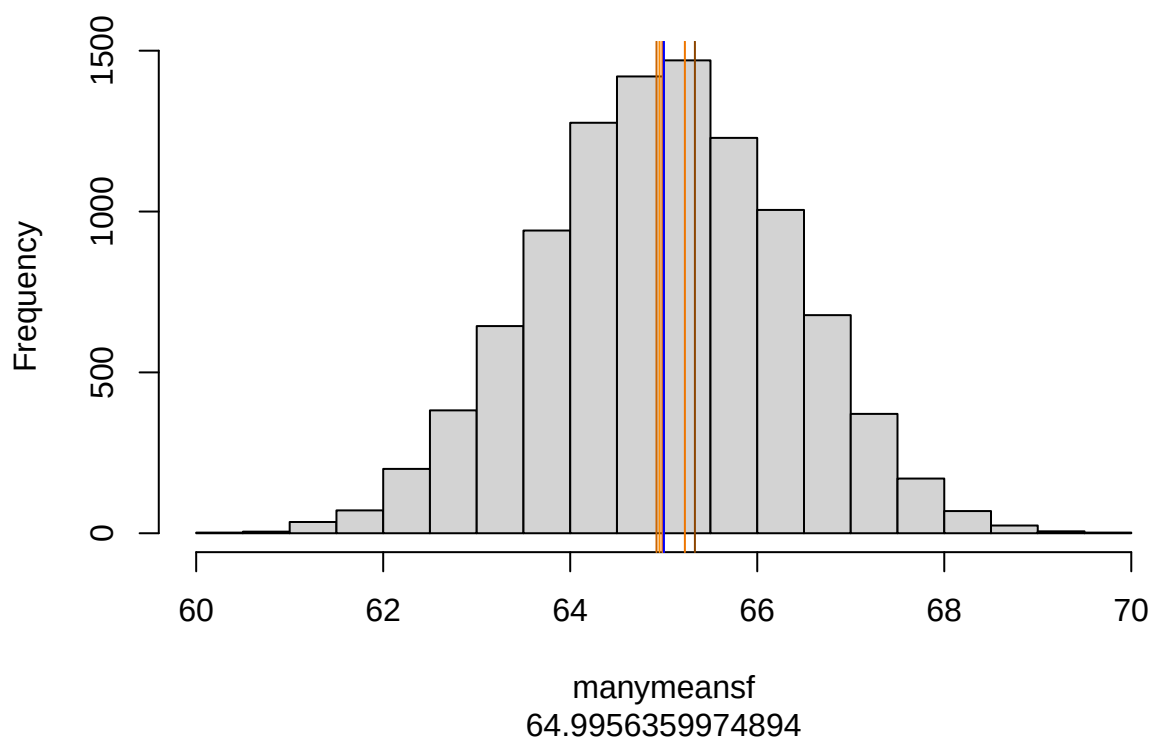
```
abline(v = mean(manymeansc), col = "darkorange2")
```

```
abline(v = mean(manymeansd), col = "darkorange1")
```

```
abline(v = mean(manymeanse), col = "darkorange")
```

```
abline(v = 65, col = "blue")
```

Many Means 10k Draws



```
mean(manymeansa)
```

```
## [1] 65.33333
```

```
mean(manymeansb)
```

```
## [1] 64.92263
```

```
mean(manymeansc)
```

```
## [1] 65.22612
```

```
mean(manymeansd)
```

```
## [1] 64.98577
```

```
mean(manymeanse)
```

```
## [1] 64.95507
```

```
mean(manymeansf) #as n increases will get closer to true mean
```

```
## [1] 64.99564
```

```
sd(manymeansa) # . . . and sd
```

```
## [1] 1.54212
```

```
sd(manymeansb)
```

```
## [1] 1.770956
```

```
sd(manymeansc)
```

```
## [1] 1.4289
```

```
sd(manymeansd)
```

```
## [1] 1.322474
```

```
sd(manymeanse)
```

```
## [1] 1.358996
```

```
sd(manymeansf)
```

```
## [1] 1.329532
```

We have two datasets today. The first looks at a measure of the concept of ideology in congress. The second is about gender and the movie industry.

As always Set your working and then Load the data(congress)

```
setwd("~/Dropbox/POLS 641/2025 641/Rcode F25/Week 4") #you will need to change for your folders  
load("congress.RData")
```

We'll start by exploring the validity and reliability of the DW-Nominate Scale

This is a measure of members of congress ideology

- A Wikipedia entry is available [here](#).)
- A longer explainer is available [here](#).
- What do you think? (in terms of validity? reliability?)

Explore the data

```
head(congress)
```

```
##   congress district   state  party      name dwnom1 dwnom2  
## 1         80         0     USA Democrat  TRUMAN -0.276  0.016  
## 2         80         1 ALABAMA Democrat  BOYKIN  F. -0.026  0.796  
## 3         80         2 ALABAMA Democrat  GRANT   G. -0.042  0.999  
## 4         80         3 ALABAMA Democrat  ANDREWS G. -0.008  1.005  
## 5         80         4 ALABAMA Democrat  HOBBS   S. -0.082  1.066  
## 6         80         5 ALABAMA Democrat  RAINS   A. -0.170  0.870
```

```
length(unique(congress$name))
```

```
## [1] 4082
```

```
## subset the data by party
```

```
rep <- subset(congress, party == "Republican") #this command uses dplyr/tidyverse
```

```
dem <- congress[congress$party == "Democrat", ] #Uses base R, but does the same thing
```

Subset Further by Congress

```
rep80 <- subset(rep, subset = (congress == 80))
```

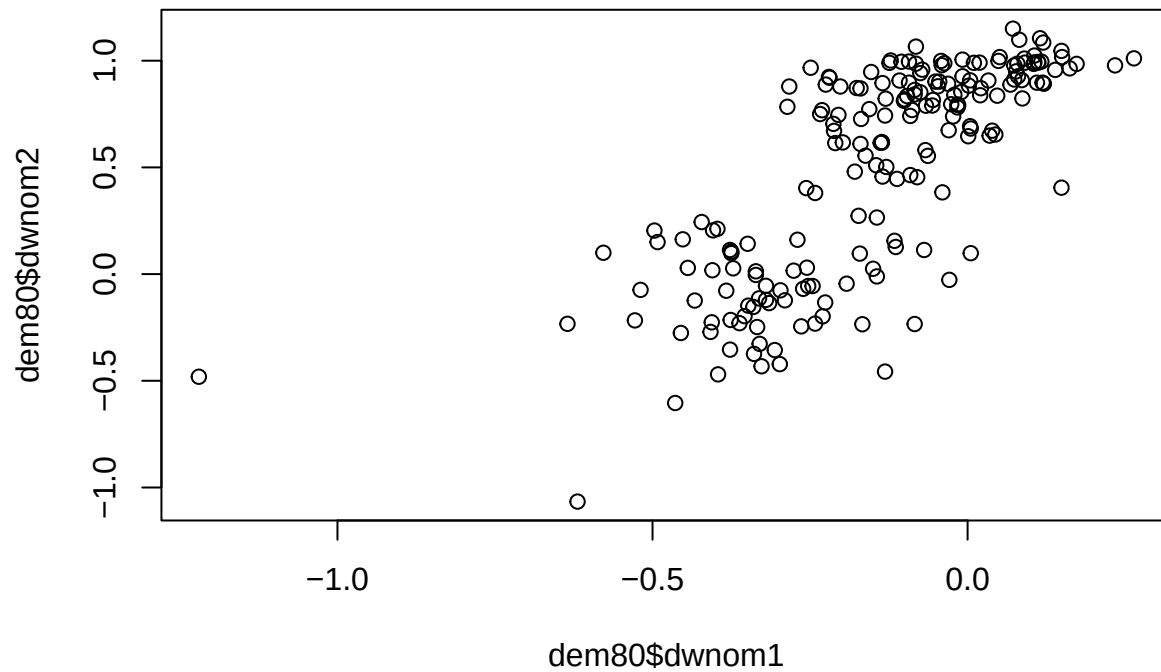
```
dem80 <- subset(dem, subset = (congress == 80))
```

```
rep112 <- subset(rep, subset = (congress == 112))
```

```
dem112 <- subset(dem, subset = (congress == 112))
```

Make a Scatterplot

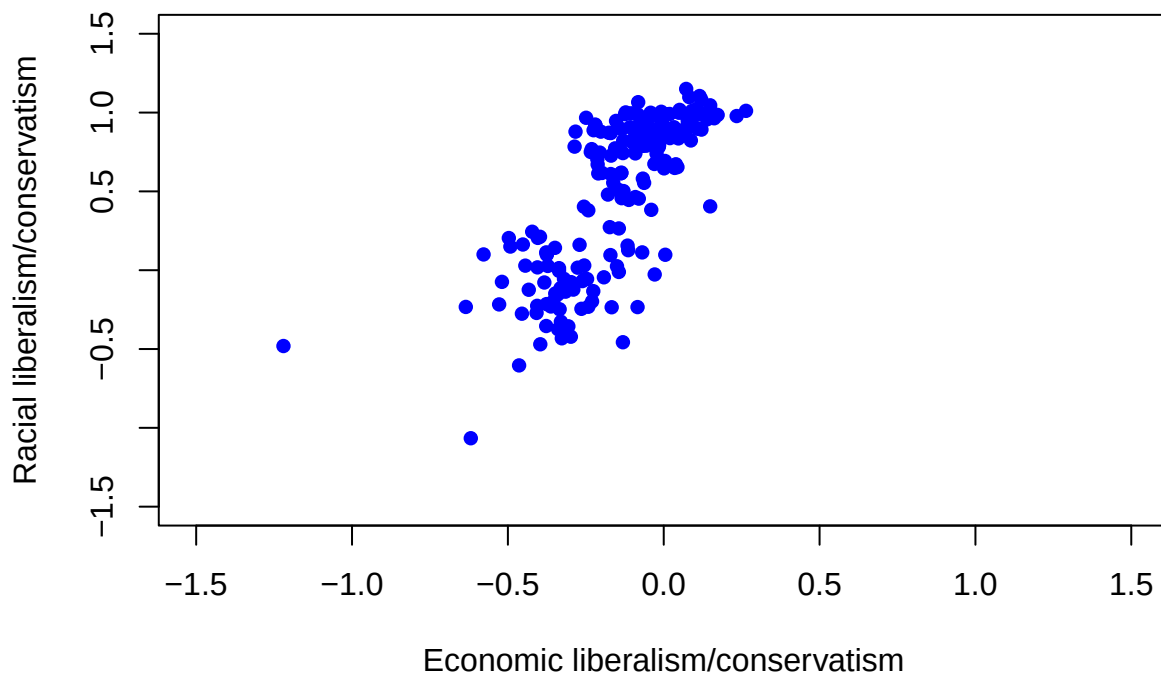
```
plot(dem80$dwnom1, dem80$dwnom2)
```

##Make it Prettier

```
Demp1ot <- plot(dem80$dwnom1, dem80$dwnom2, pch = 16, col = "blue", xlim = c(-1.5, 1.5), ylim = c(-1.5, 1.5), xlab = "Economic liberalism/conservatism", ylab = "Racial liberalism/conservatism", main = "80th Congress")
```

80th Congress



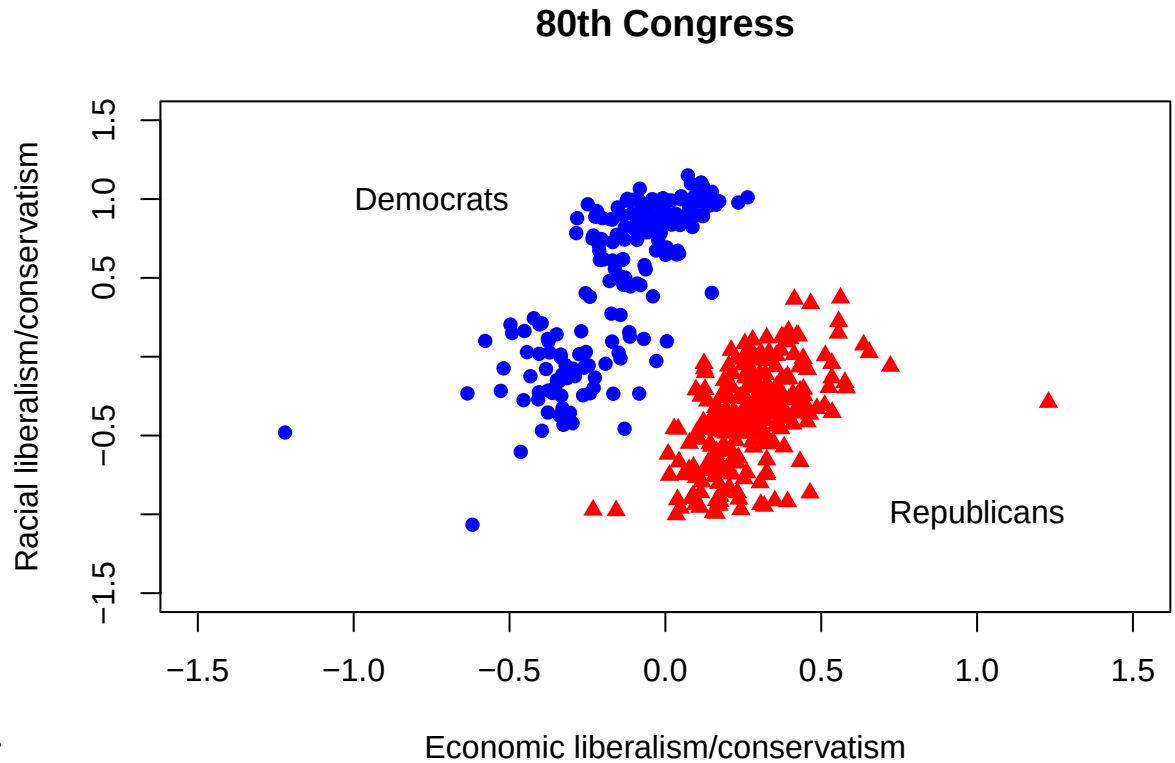
does this tell us about the two dimensions?

What

##Let's add the GOP

```
plot(dem80$dwnom1, dem80$dwnom2, pch = 16, col = "blue", xlim = c(-1.5, 1.5), ylim = c(-1.5, 1.5), xlab = "Economic liberalism/conservatism", ylab = "Racial liberalism/conservatism", main = "80th Congress")
points(rep80$dwnom1, rep80$dwnom2, pch = 17, col = "red")
text(-0.75, 1, "Democrats") #adding a label to our plot
text(1, -1, "Republicans")
```

When plotting, you have to send ALL the commands at once



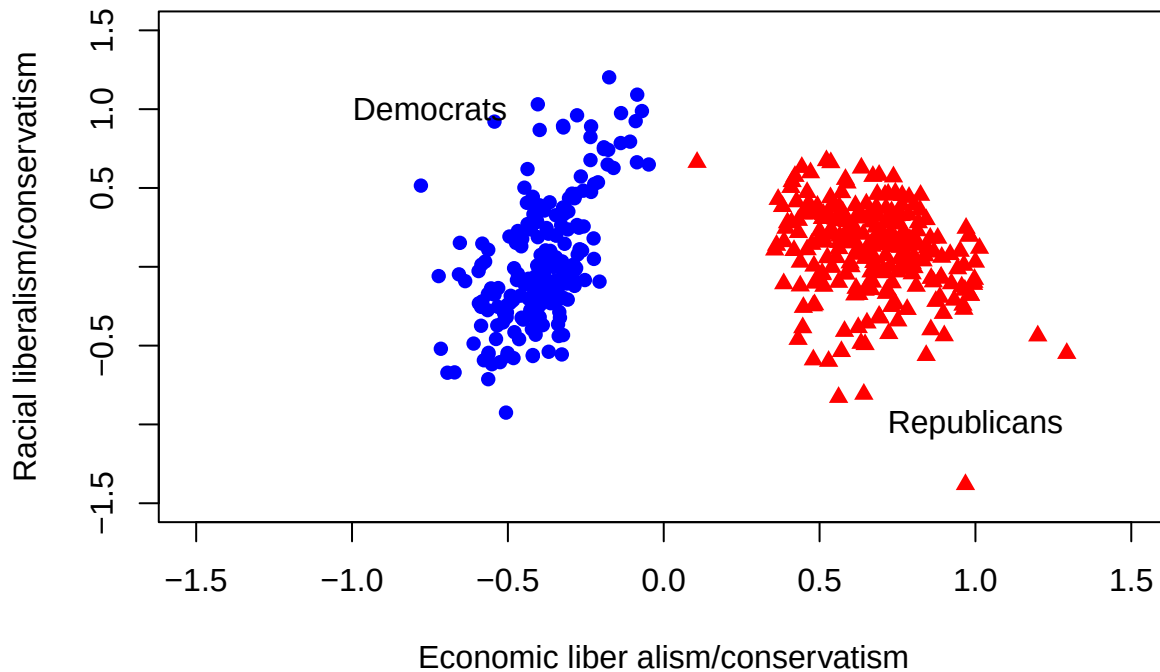
##Ain't it Perty!?

##Class Exercise

Can you produce an identical plot for the 112th Congress?

What do these plots show us? Does the measure have face validity?

112th Congress



##Let's Dig into this Further

How have the median R and D Members of Congress changed over time?

First calculate the median for each party by Congress:

```
dem.median <- tapply(dem$dwnom1, dem$congress, median) #Find the median of dwnom1 for each congress. Y
rep.median <- tapply(rep$dwnom1, rep$congress, median)
```

#tapply is a useful command, but you won't need it for problem sets etc. Dplyr group_by does similar things

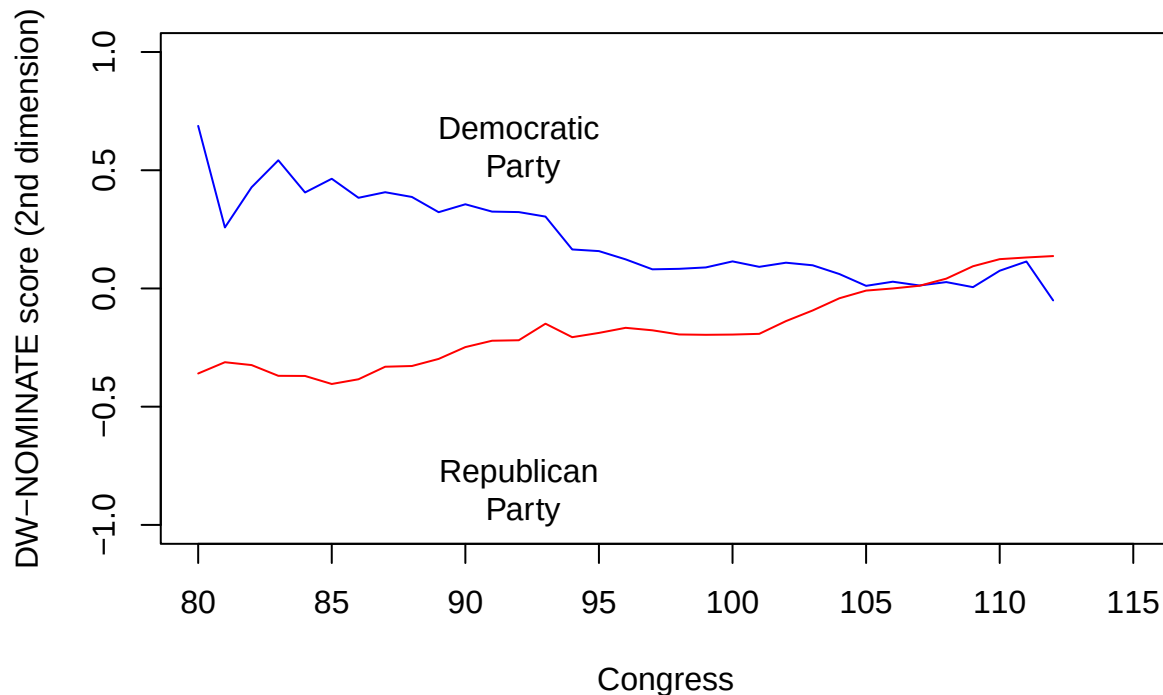
Now let's plot those medians for each party over time:

```
plot(names(dem.median), dem.median, col = "blue", type = "l", xlim = c(80, 115),
     ylim = c(-1, 1), xlab = "Congress", ylab = "DW-NOMINATE score (1st dimension)")
lines(names(rep.median), rep.median, col = "red")
text(110, -0.6, "Democratic\n Party")
text(110, 0.85, "Republican\n Party")
```

#Now repeat for the 2nd Dimension

```
dem.median2 <- tapply(dem$dwnom2, dem$congress, median) #Find the median of dwnom1 for each congress.
rep.median2 <- tapply(rep$dwnom2, rep$congress, median)
```

```
plot(names(dem.median2), dem.median2, col = "blue", type = "l", xlim = c(80, 115),
     ylim = c(-1, 1), xlab = "Congress", ylab = "DW-NOMINATE score (2nd dimension)")
lines(names(rep.median2), rep.median2, col = "red")
text(92, 0.6, "Democratic\n Party")
text(92, -0.85, "Republican\n Party")
```



Bonus:

Why might we want to use the median instead of the mean here? Does using the mean change the broader story?

Really interesting viewing of DW Nominate data: <https://www.youtube.com/watch?v=M193ZIOFk0k>

Load the data: These data comes from this fivethirtyeight article

```
load("bech.RData")
```

Load Packages: we'll use ggplot briefly again today

```
library(ggplot2)
```

Find the number of variables and observations - *What is our unit of analysis?*

```
head(bech)
```

```
##   year      imdb      title      test clean_test binary    budget
## 1 2013 tt1711425    21 & Over    notalk    notalk   FAIL 13000000
## 2 2012 tt1343727    Dredd 3D  ok-disagree    ok    PASS 45000000
## 3 2013 tt2024544 12 Years a Slave notalk-disagree    notalk   FAIL 20000000
## 4 2013 tt1272878      2 Guns    notalk    notalk   FAIL 61000000
## 5 2013 tt0453562      42      men      men   FAIL 40000000
## 6 2013 tt1335975    47 Ronin      men      men   FAIL 22500000
##  domgross  intgross    code budget_2013 domgross_2013 intgross_2013
## 1 25682380 42195766 2013FAIL 13000000 25682380 42195766
## 2 13414714 40868994 2012PASS 45658735 13611086 41467257
## 3 53107035 158607035 2013FAIL 20000000 53107035 158607035
## 4 75612460 132493015 2013FAIL 61000000 75612460 132493015
## 5 95020213 95020213 2013FAIL 40000000 95020213 95020213
## 6 38362475 145803842 2013FAIL 225000000 38362475 145803842
```

VARIABLE NAMES

```
names(bech)
```

```
## [1] "year"      "imdb"      "title"     "test"
```

```
## [5] "clean_test"      "binary"          "budget"          "domgross"
## [9] "intgross"        "code"            "budget_2013"     "domgross_2013"
## [13] "intgross_2013"
```

```
variable.names(bech)
```

```
## [1] "year"           "imdb"           "title"          "test"
## [5] "clean_test"     "binary"         "budget"         "domgross"
## [9] "intgross"       "code"           "budget_2013"    "domgross_2013"
## [13] "intgross_2013"
```

What time range does the data cover?

```
min(bech$year)
```

```
## [1] 1970
```

```
max(bech$year)
```

```
## [1] 2013
```

```
range(bech$year)
```

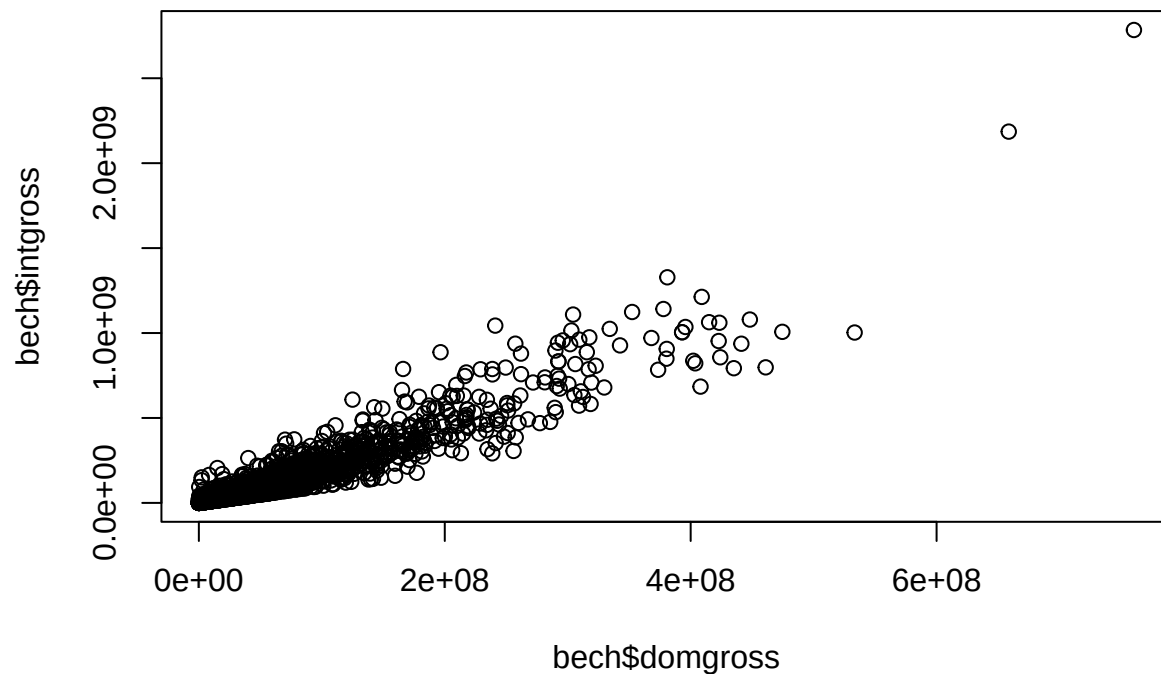
```
## [1] 1970 2013
```

Exploring the data

```
table(bech$clean_test)
```

```
##
## dubious      men  notalk nowomen      ok
##      141      193      510      138     794
```

```
plot(bech$domgross, bech$intgross)
```



```
prop.table(table(bech$binary))
```

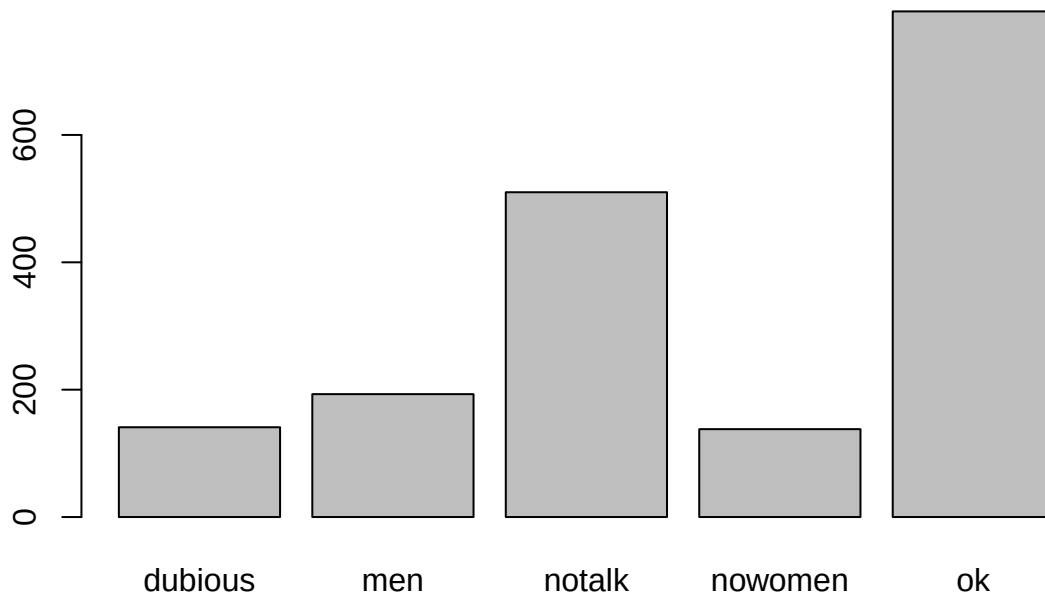
```
##
```

```
##      FAIL      PASS
## 0.5529279 0.4470721
```

```
prop.table(table(bech$clean_test, bech$binary))
```

```
##
##      FAIL      PASS
## dubious 0.07939189 0.00000000
## men     0.10867117 0.00000000
## notalk  0.28716216 0.00000000
## nowomen 0.07770270 0.00000000
## ok      0.00000000 0.44707207
```

```
barplot(table(bech$clean_test)) #show ugly bar plot
```



FIND THE RANGE OF GROSS REVENUE(we'll just look at domestic and standardize to 2013 dollar value)

```
range(bech$domgross_2013)
```

```
## [1]      899 1771682790
```

```
max(bech$domgross_2013) #Bonus: What movie is this? Hint: View(yourdata)
```

```
## [1] 1771682790
```

```
min(bech$domgross_2013) #Cheers to you if you've actually watched this
```

```
## [1] 899
```

FIND STANDARD DEVIATION AND VARIANCE - We'll take a closer look at this next week. Formula for Standard Deviation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

More info on using math symbols in Rmarkdown is here: <https://rpruim.github.io/s341/S19/from-class/MathinRmd.html>

```
sqrt(sum((bech$domgross_2013 - mean(bech$domgross_2013))^2/(length(bech$domgross_2013) - 1))) #How would we turn this into the variance?
```

```
## [1] 125965349
```

Of course, it is much easier to just have R do it

```
sd(bech$domgross_2013)
```

```
## [1] 125965349
```

```
var(bech$domgross_2013)
```

```
## [1] 1.586727e+16
```

```
sd(bech$domgross_2013)^2 #Variance = standard deviation squared.
```

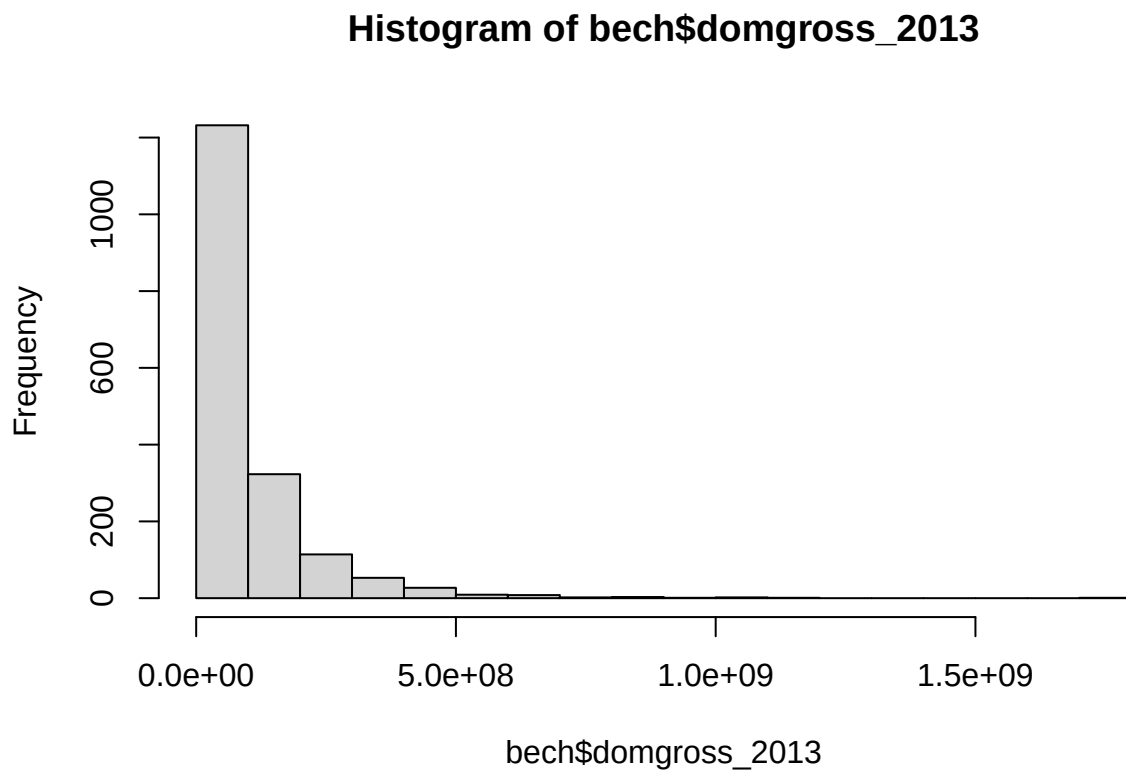
```
## [1] 1.586727e+16
```

```
sqrt(var(bech$domgross_2013)) #Standard deviation = square root of variance.
```

```
## [1] 125965349
```

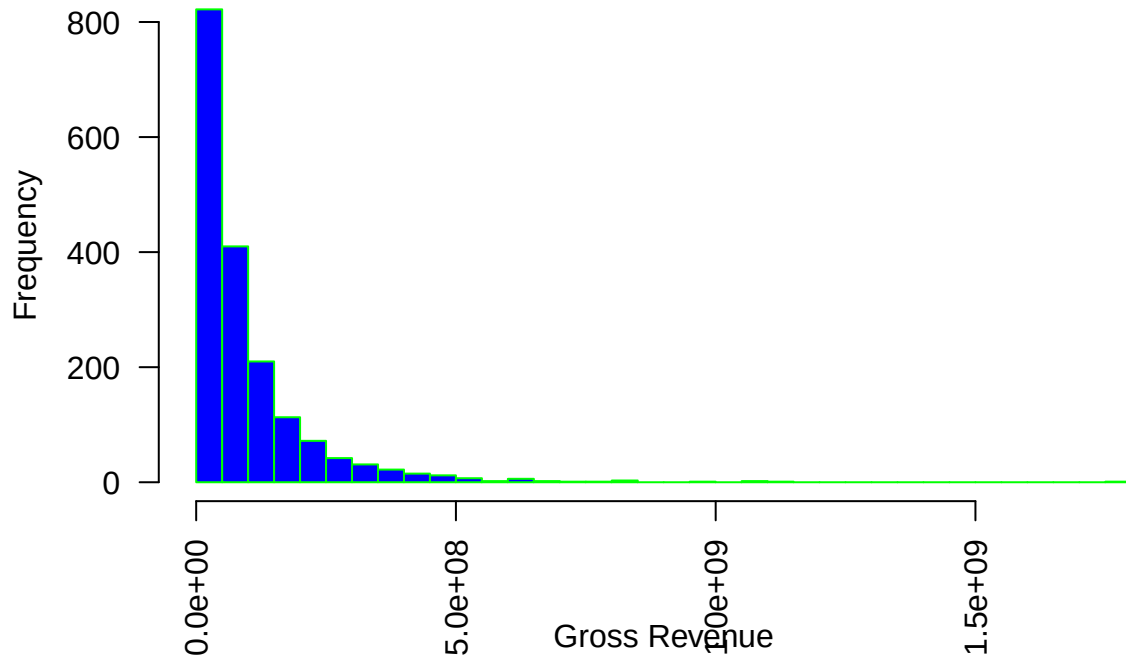
TAKE A LOOK AT THE DISTRIBUTION OF DOMESTIC GROSS

```
hist(bech$domgross_2013, breaks = 15) #breaks= number of bars
```



```
hist(bech$domgross_2013, main = "Distribution of Gross Revenue", xlab = "Gross Revenue",  
col = "blue", border = "green", breaks = 30, las = 2) #bedazzled
```

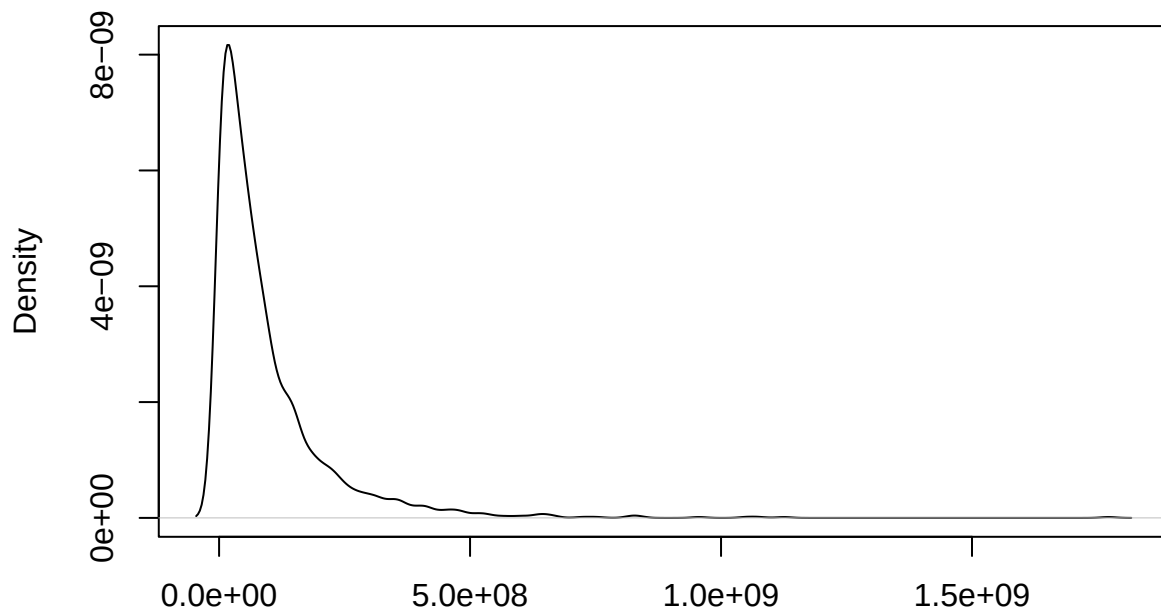
Distribution of Gross Revenue



LOOK AT THE DENSITY PLOT OF DOMESTIC GROSS

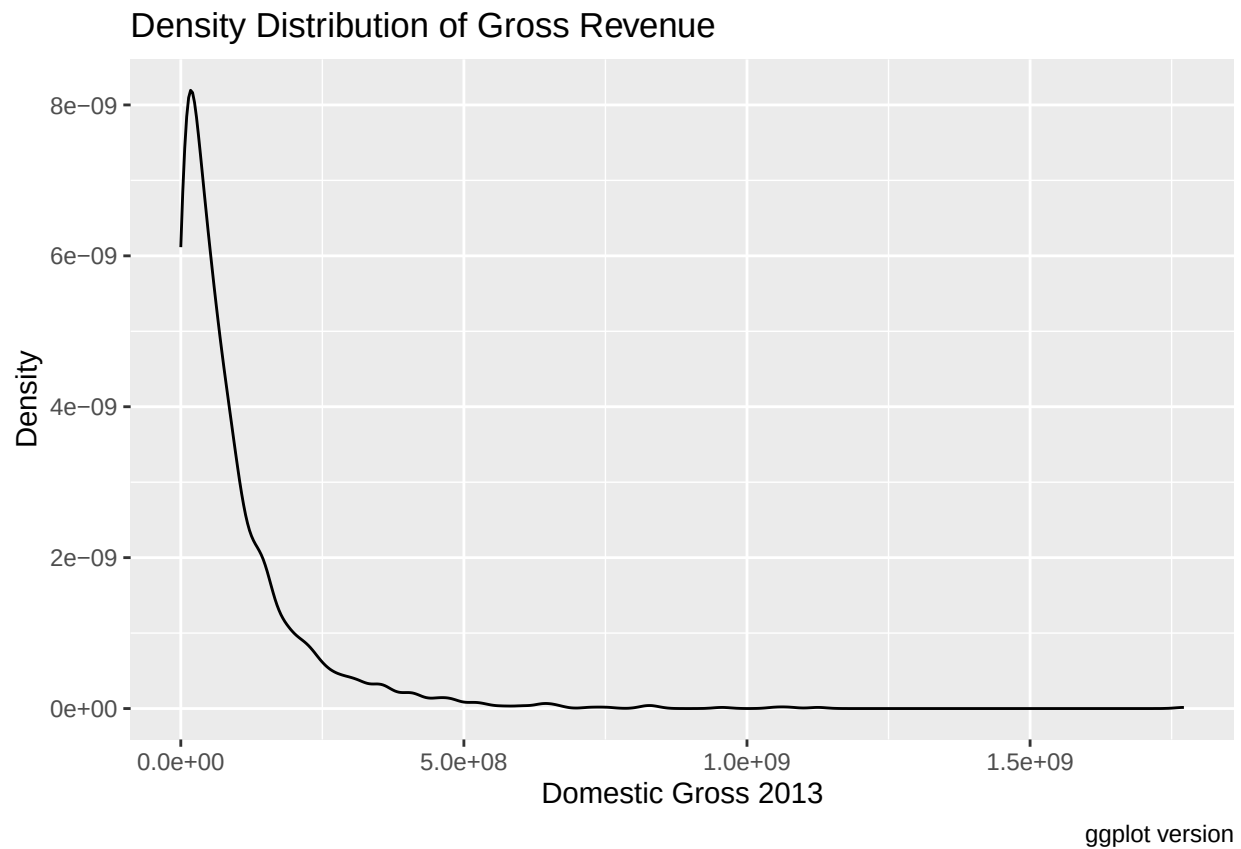
```
plot(density(bech$domgross_2013), main = "Density Distribution of Gross Revenue")
```

Density Distribution of Gross Revenue



N = 1776 Bandwidth = 1.521e+07

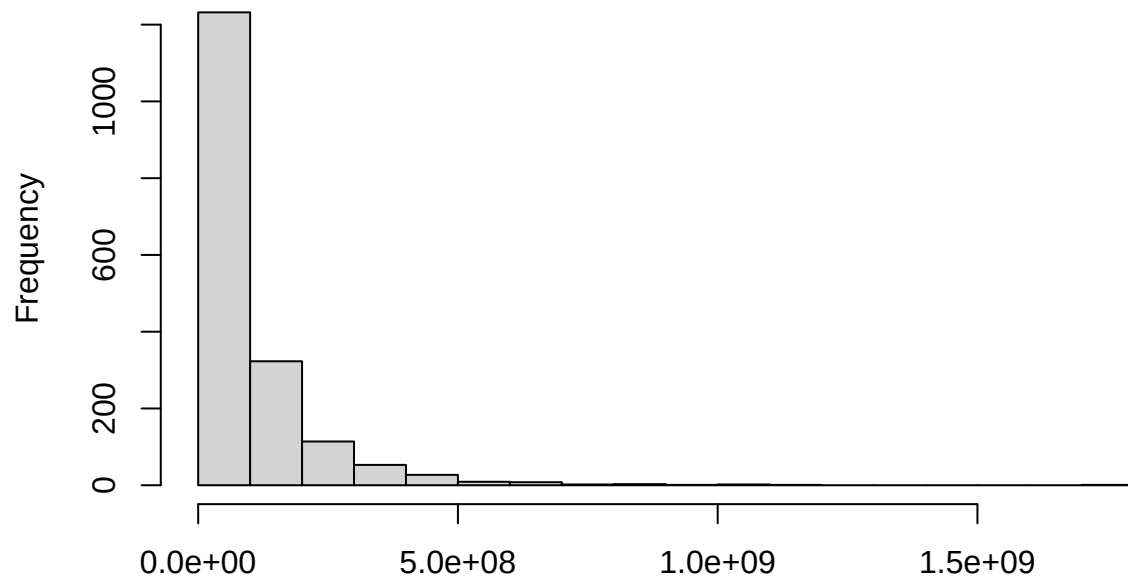
```
ggplot(bech, aes(x = domgross_2013)) + geom_density() + labs(x = "Domestic Gross 2013",  
y = "Density", title = "Density Distribution of Gross Revenue", caption = "ggplot version")
```

How are histogram and density plot different? Which is closer to a pdf?

```
hist(bech$domgross_2013, main = "Histogram of of Gross Revenue", xlab = "Domestic Gross 2013",  
     breaks = 20)
```

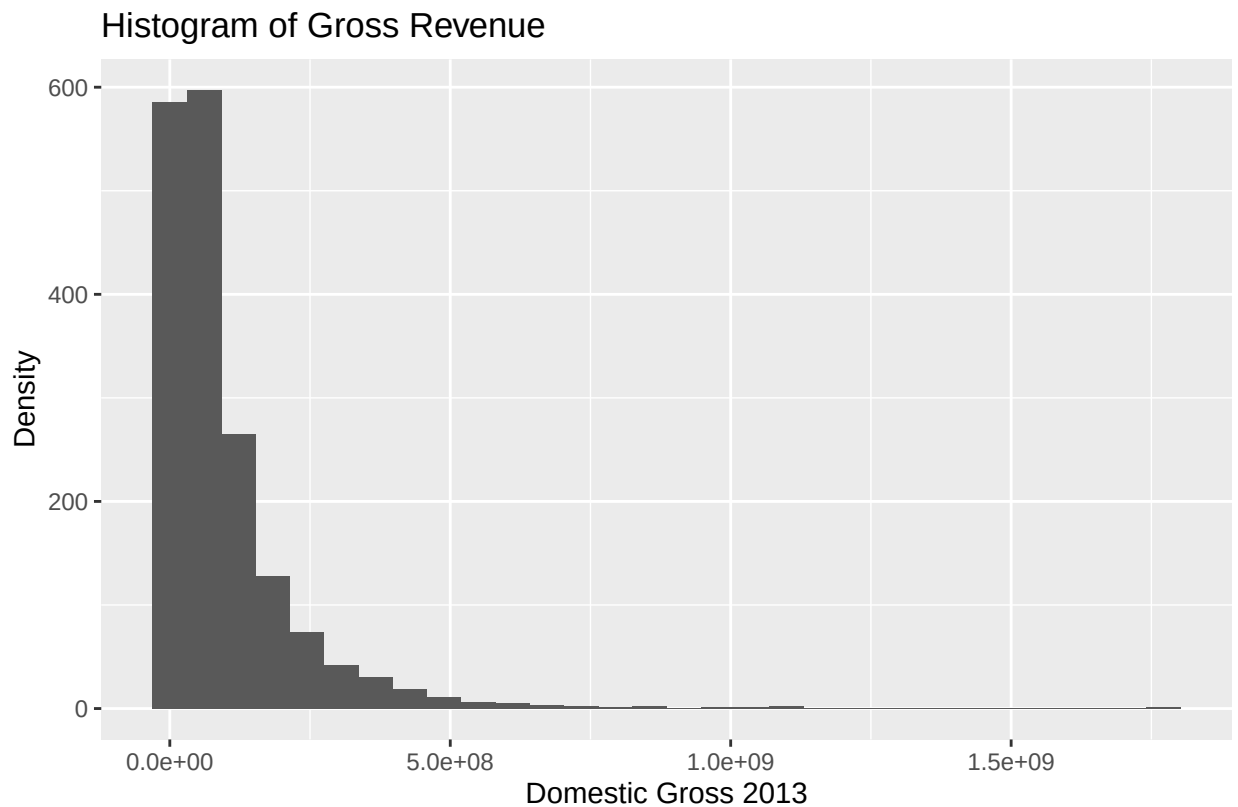
Histogram of of Gross Revenue



Domestic Gross 2013

```
ggplot(bech, aes(x = domgross_2013)) + geom_histogram() + labs(x = "Domestic Gross 2013",  
  y = "Density", title = "Histogram of Gross Revenue", caption = "ggplot version")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

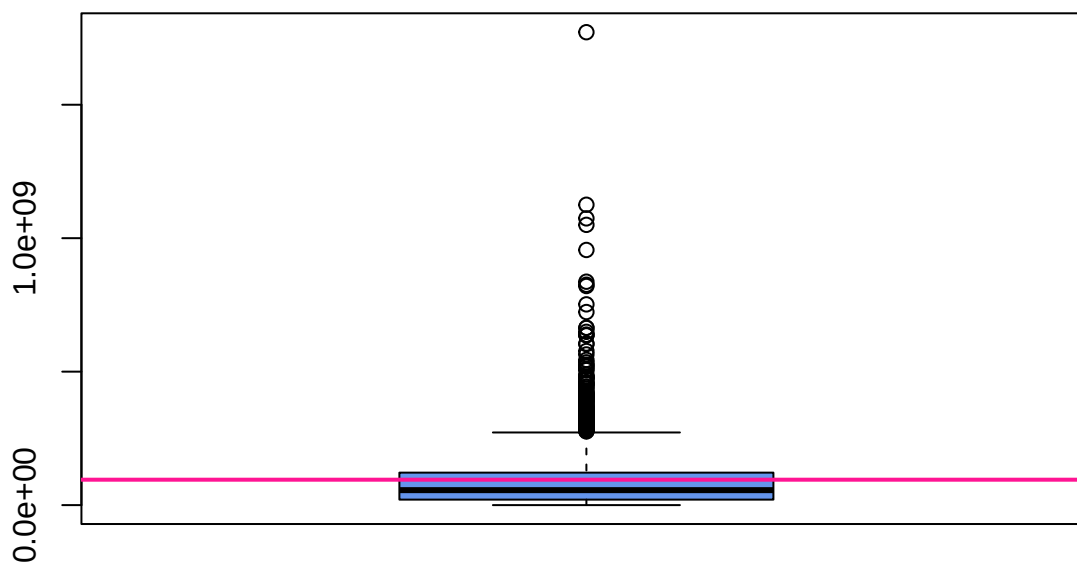


ggplot version

MAKE A BOXPLOT

```
boxplot(bech$domgross_2013, main = "Boxplot of Gross Revenue", col = "cornflowerblue")
abline(h = mean(bech$domgross_2013), col = "deeppink", lwd = 2)
```

Boxplot of Gross Revenue



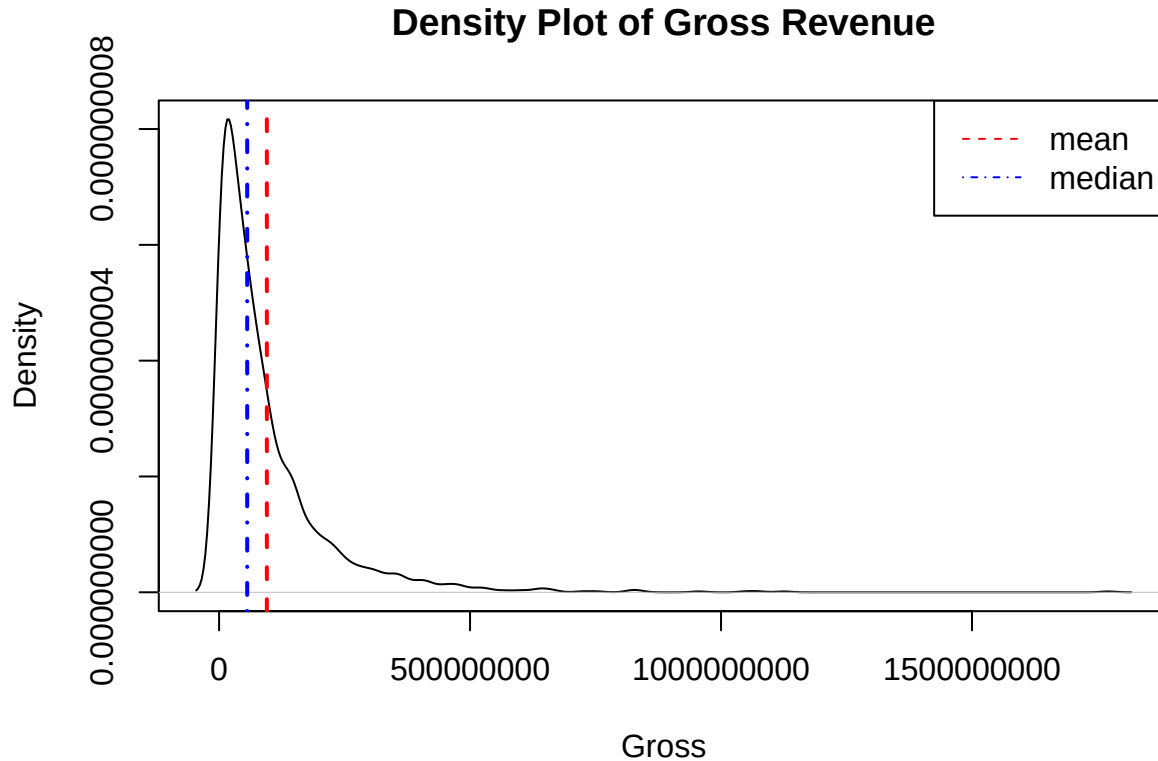
A boxplot

shows the median, 25% and 75% quartile, and important outliers

ADD LINES TO INDICATE THE MEAN AND MEDIAN, AND MAKE A LEGEND (*Make sure all the*

code stays together)

```
plot(density(bech$domgross_2013), main = "Density Plot of Gross Revenue", xlab = "Gross")
abline(v = mean(bech$domgross_2013), lty = 2, lwd = 2, col = "red") # put a vertical line at the mean
abline(v = median(bech$domgross_2013), lty = 4, lwd = 2, col = "blue")
legend("topright", legend = c("mean", "median"), lty = c(2, 4), col = c("red", "blue"))
```

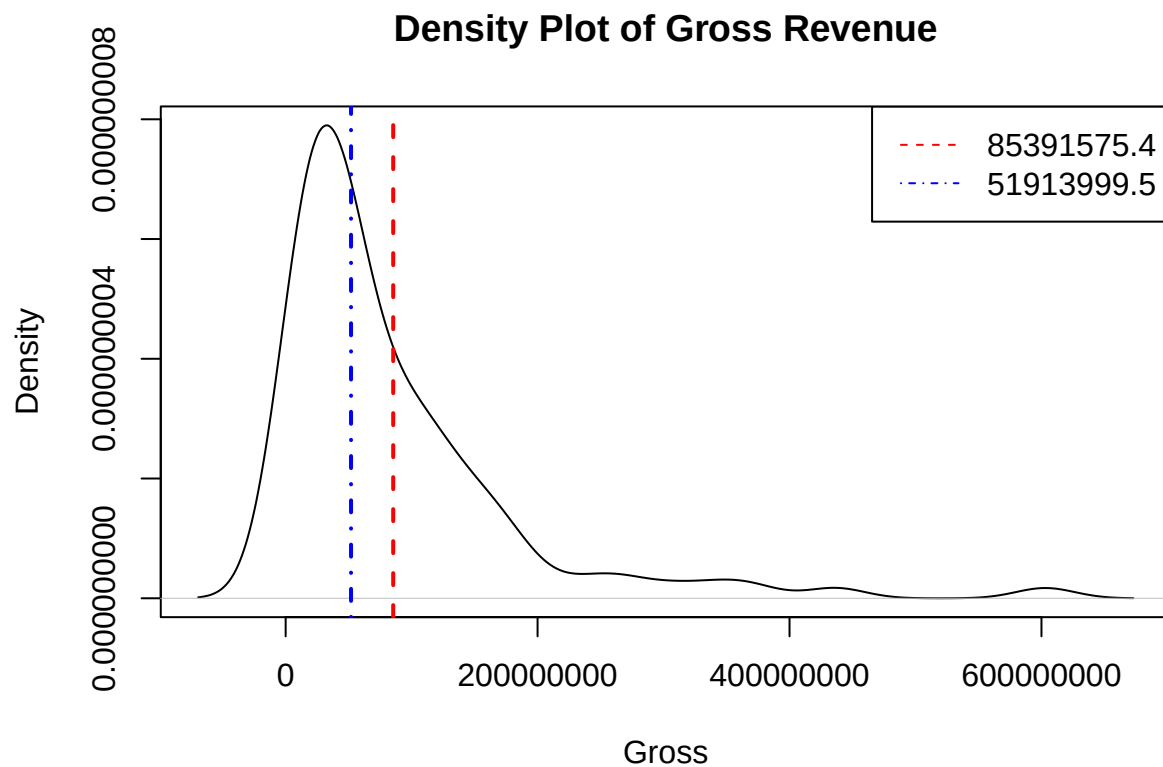


```
options(scipen = 10) #This stops use of scientific notation
```

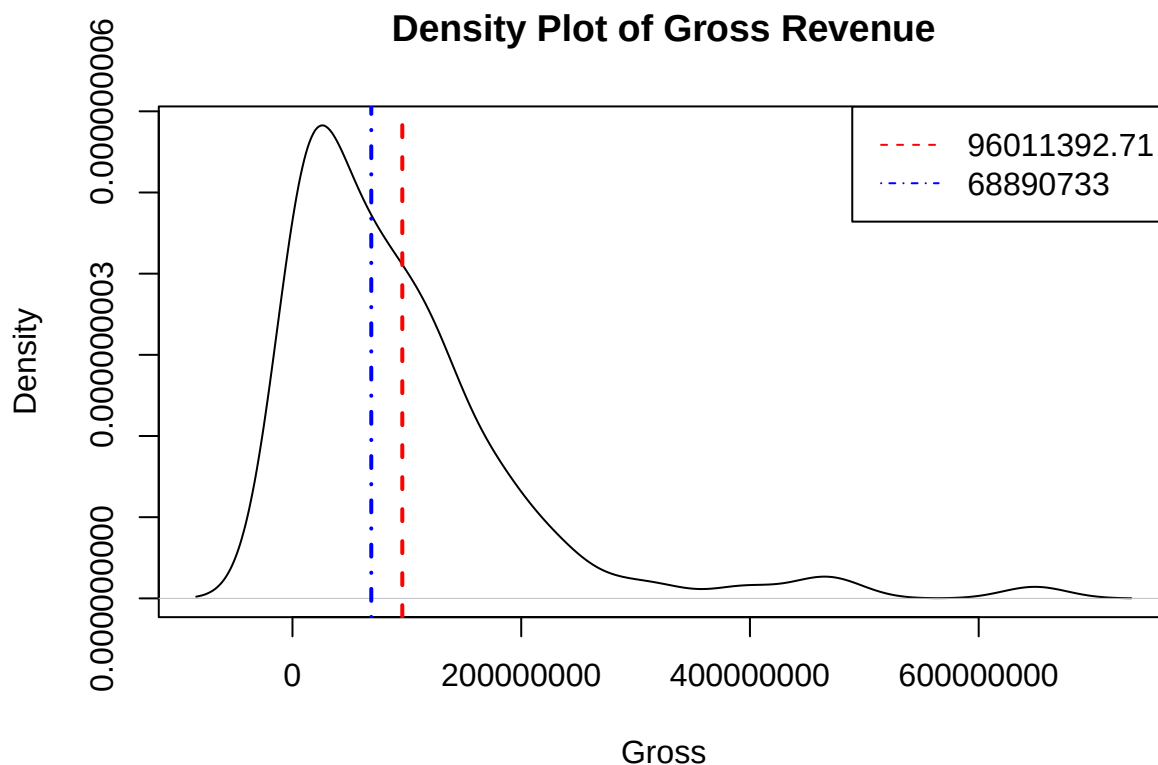
Take Random Samples and see how mean and median changes

```
sample1 <- bech[sample(nrow(bech), 100), ] #Randomly samples 100 movies

plot(density(sample1$domgross_2013), main = "Density Plot of Gross Revenue", xlab = "Gross")
abline(v = mean(sample1$domgross_2013), lty = 2, lwd = 2, col = "red")
abline(v = median(sample1$domgross_2013), lty = 4, lwd = 2, col = "blue")
legend("topright", legend = c(mean(sample1$domgross_2013), median(sample1$domgross_2013)),
      lty = c(2, 4), col = c("red", "blue"))
```



```
sample2 <- bech[sample(nrow(bech), 100), ]
plot(density(sample2$domgross_2013), main = "Density Plot of Gross Revenue", xlab = "Gross")
abline(v = mean(sample2$domgross_2013), lty = 2, lwd = 2, col = "red")
abline(v = median(sample2$domgross_2013), lty = 4, lwd = 2, col = "blue")
legend("topright", legend = c(mean(sample2$domgross_2013), median(sample2$domgross_2013)),
      lty = c(2, 4), col = c("red", "blue"))
```



```
mean(sample1$domgross_2013)
```

```
## [1] 85391575
```

```
mean(sample2$domgross_2013)
```

```
## [1] 96011393
```

```
median(sample1$domgross_2013)
```

```
## [1] 51914000
```

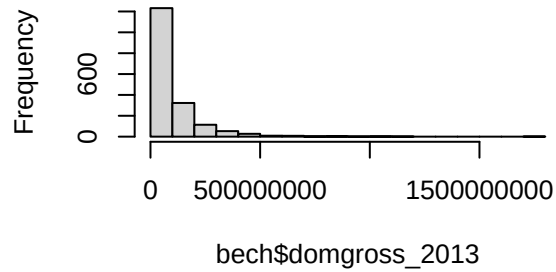
```
median(sample2$domgross_2013)
```

```
## [1] 68890733
```

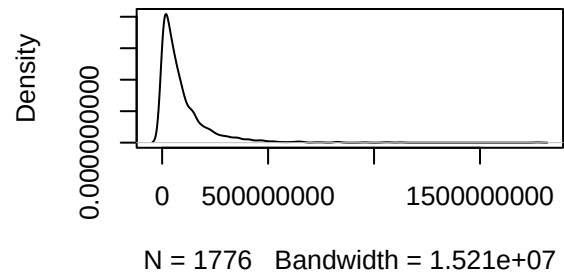
LOOK AT THE SAMPLING DISTRIBUTION OF REVENUE AND BUDGET (*Let's put the four figures arranged in 2 rows and 2 columns*)

```
par(mfrow = c(2, 2)) ###This line of code tells R you want a 2 x 2 of smaller graphs
hist(bech$domgross_2013, main = "Histogram of Domestic Gross", breaks = 15)
plot(density(bech$domgross_2013), main = "Distribution of Domestic Gross")
hist(bech$budget_2013, main = "Histogram of Budget", breaks = 15)
plot(density(bech$budget_2013), main = "Distribution of Budget")
```

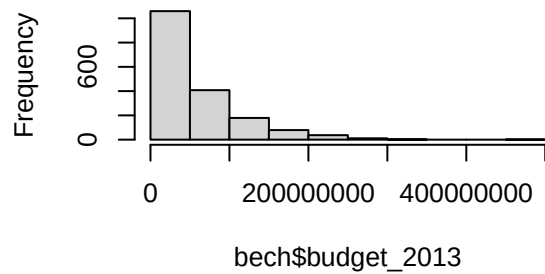
Histogram of Domestic Gross



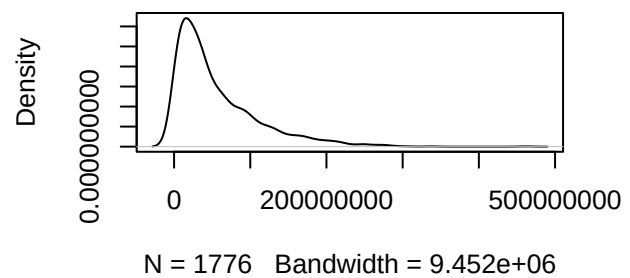
Distribution of Domestic Gross



Histogram of Budget



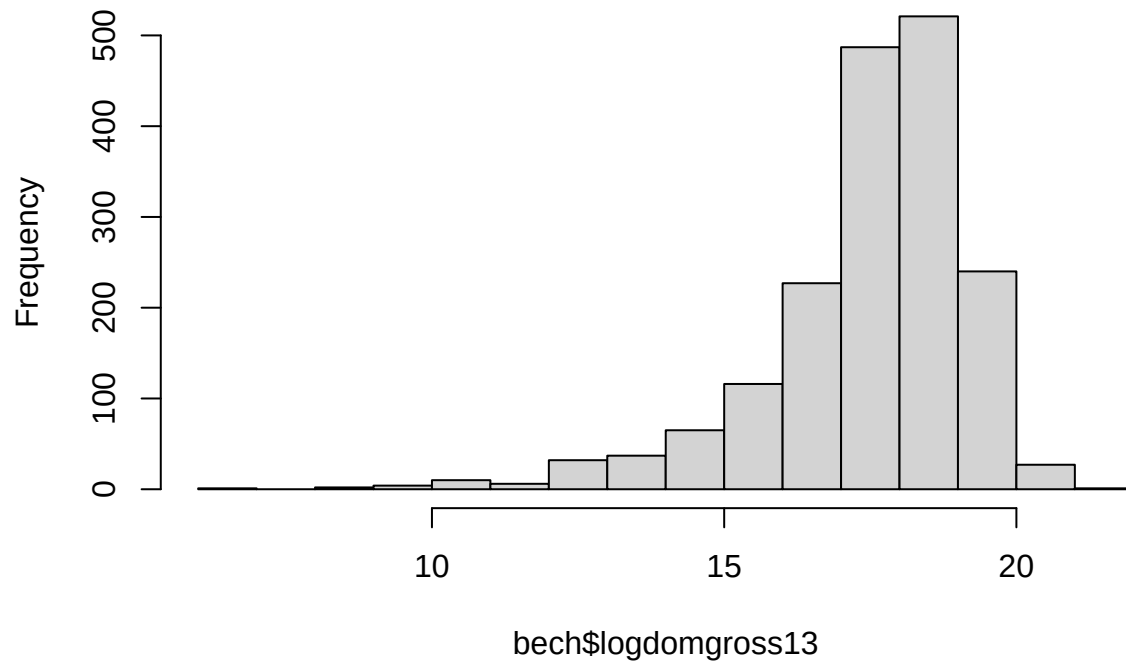
Distribution of Budget



THIS IS NOT A NORMAL DISTRIBUTION SO MAY WANT TO CREATE LOGGED VERSION We will talk about transforming variables later

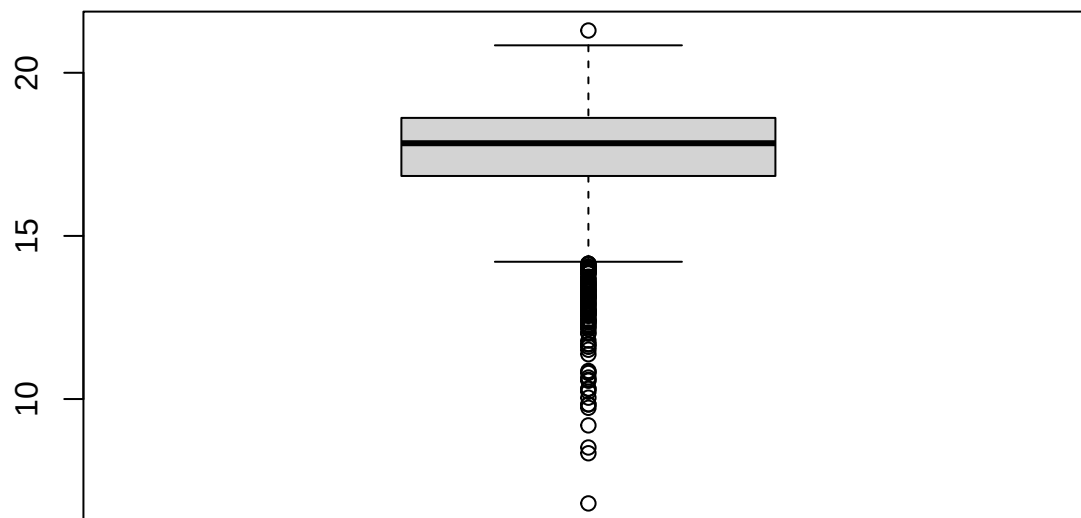
```
par(mfrow = c(1, 1)) #Return to just looking at one graph at a time
bech$logdomgross13 <- log(bech$domgross_2013) #creates new logged variable
hist(bech$logdomgross13, main = "Logged Domestic Revenue") #Looks more normal, harder to interpret
```

Logged Domestic Revenue



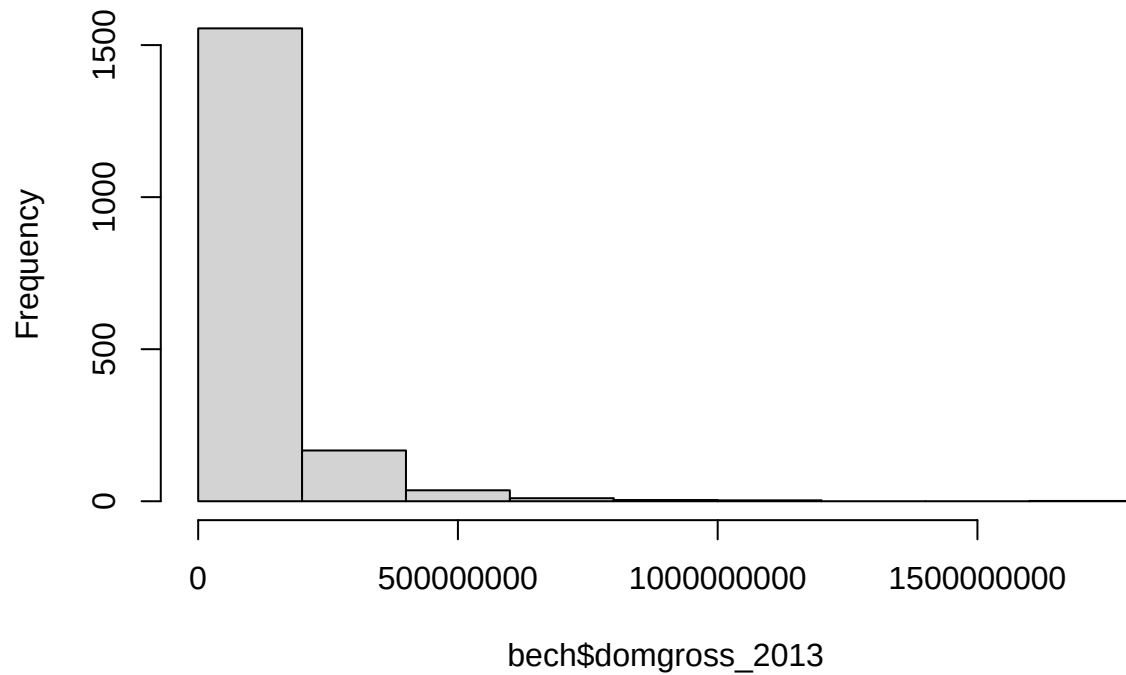
```
boxplot(bech$logdomgross13, main = "Logged Domestic Revenue")
```

Logged Domestic Revenue



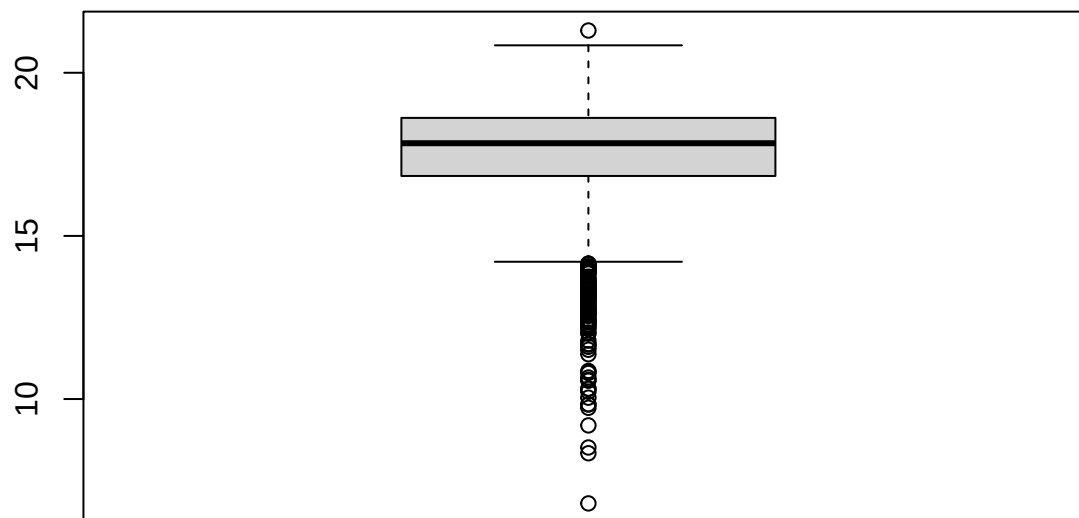
```
hist(bech$domgross_2013, main = "Domestic Revenue") #Looks more normal, harder to interpret
```


Domestic Revenue



```
boxplot(bech$logdomgross13, main = "Logged Domestic Revenue")
```

Logged Domestic Revenue



LET'S USE STATS TO MAKE COMPARISONS

First, find averages for films that pass and fail the Bechdel test (there are several ways to do this).

```
pass <- bech[which(bech$binary == "PASS"), ] #Creates a new dataset of only those that pass the bechdel test
fail <- bech[which(bech$binary == "FAIL"), ]
mean(pass$domgross_2013)
```

```
## [1] 79591919
```

```
mean(fail$domgross_2013)
```

```
## [1] 107774371
```

```
# How many films passed and how many failed?
```

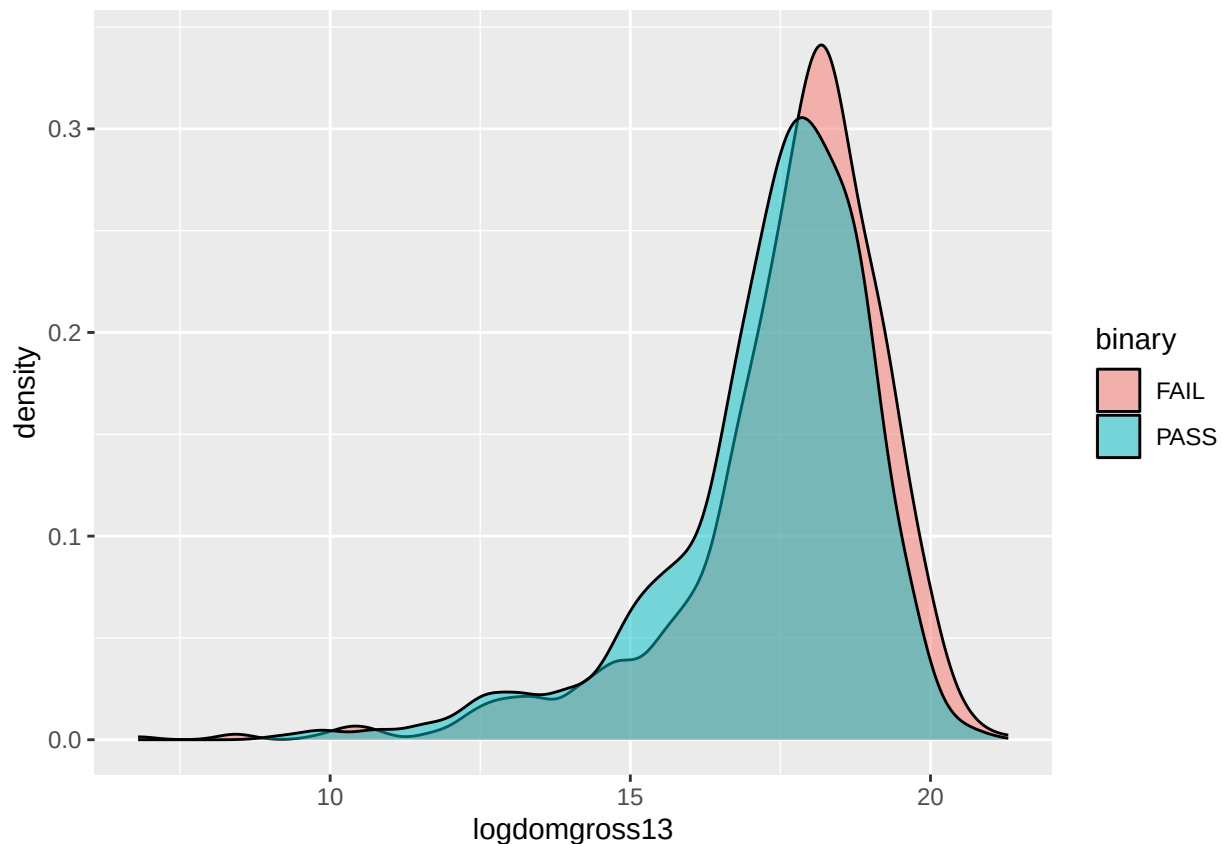
```
mean(pass$budget_2013)
```

```
## [1] 46670698
```

```
mean(fail$budget_2013)
```

```
## [1] 63343993
```

```
ggplot(bech, aes(x = logdomgross13, fill = binary)) + geom_density(alpha = 0.5, position = "identity")
```



Second, simple algebra to find difference.

```
grossdiff <- mean(pass$domgross_2013) - mean(fail$domgross_2013)  
grossdiff
```

```
## [1] -28182452
```

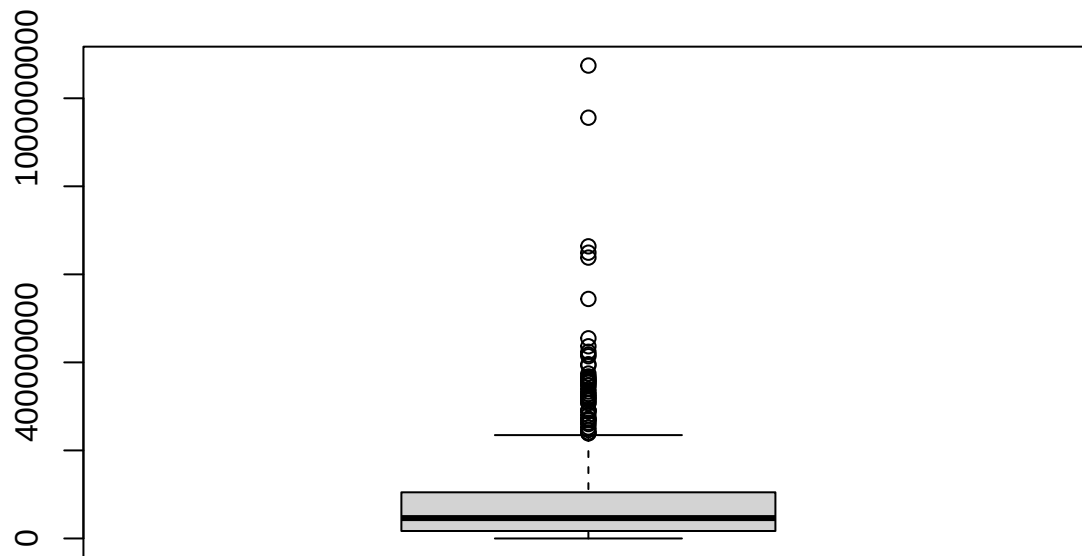
Another method for finding mean of subset (more here: <http://www.statmethods.net/management/subset.ht> ml)

```
xbar_pass_t <- mean(bech$domgross_2013[bech$binary == "PASS"])  
xbar_fail_t <- mean(bech$domgross_2013[bech$binary == "FAIL"])  
DIFF <- (xbar_pass_t - xbar_fail_t)  
DIFF
```

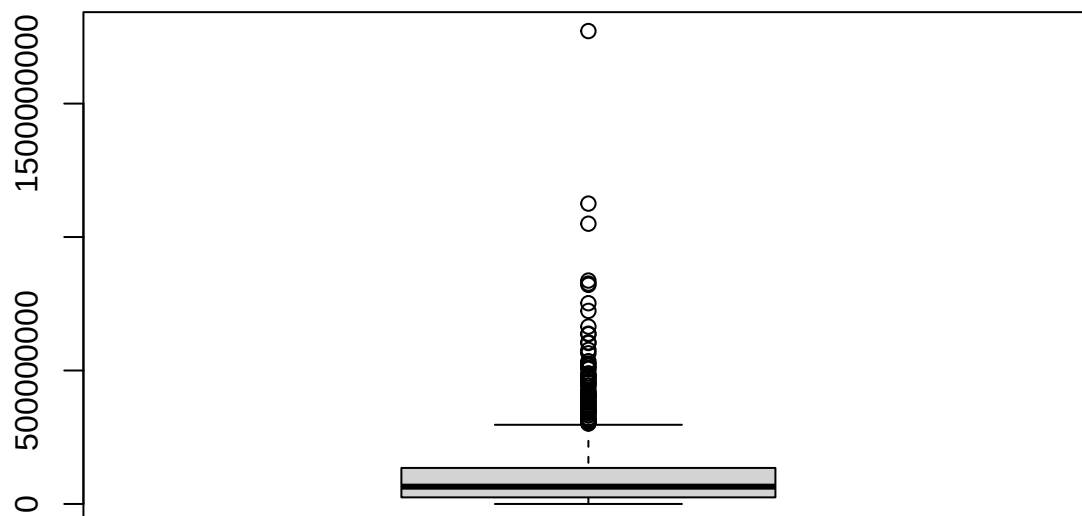
```
## [1] -28182452
```

Show Both Box Plots

```
boxplot(pass$domgross_2013)
```

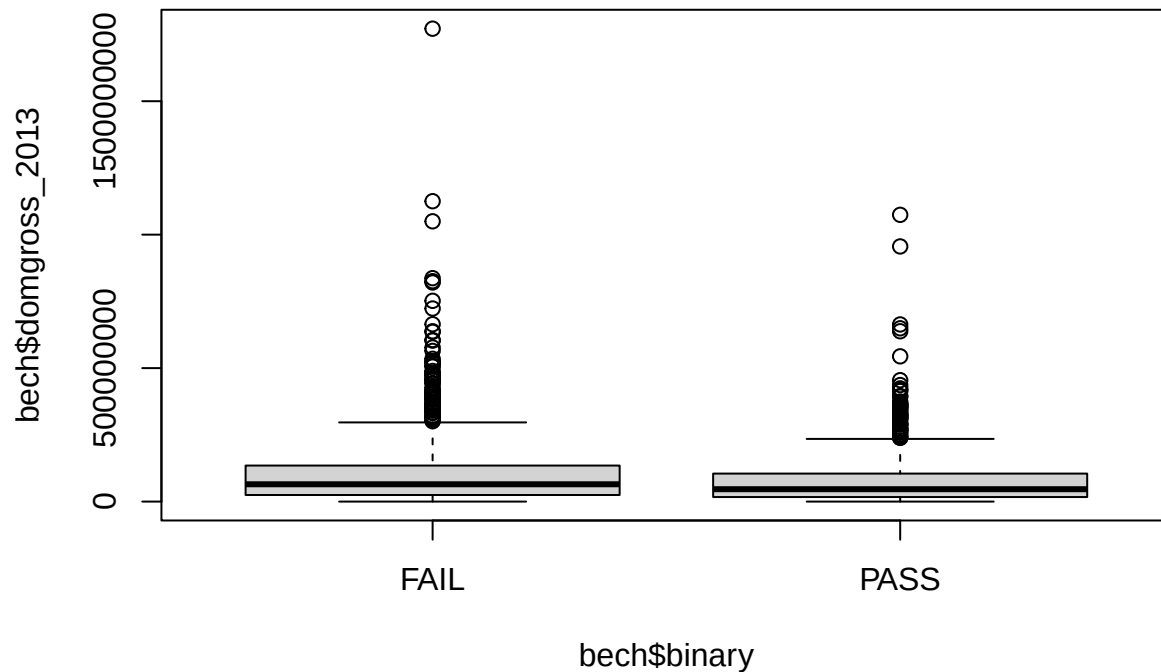


```
boxplot(fail$domgross_2013)
```

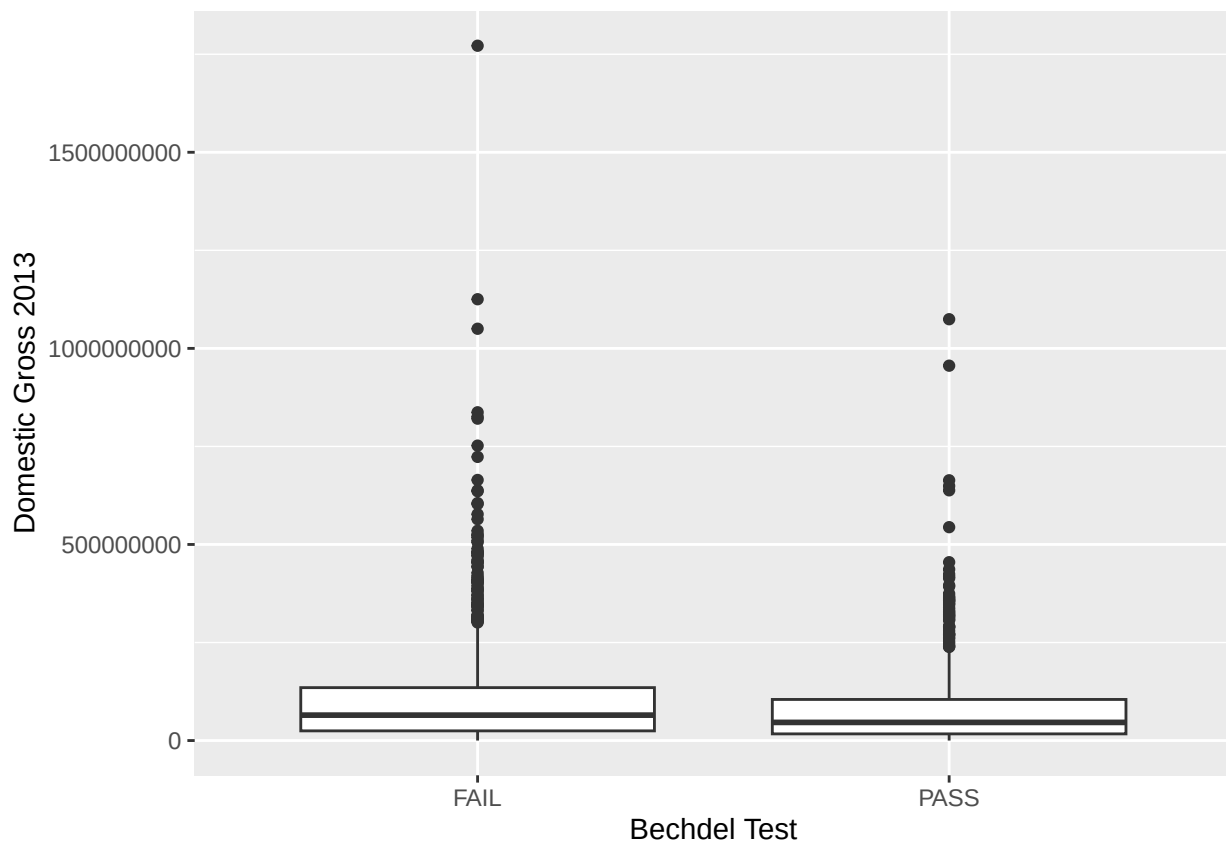


or from original dataset

```
boxplot(bech$domgross_2013 ~ bech$binary)
```

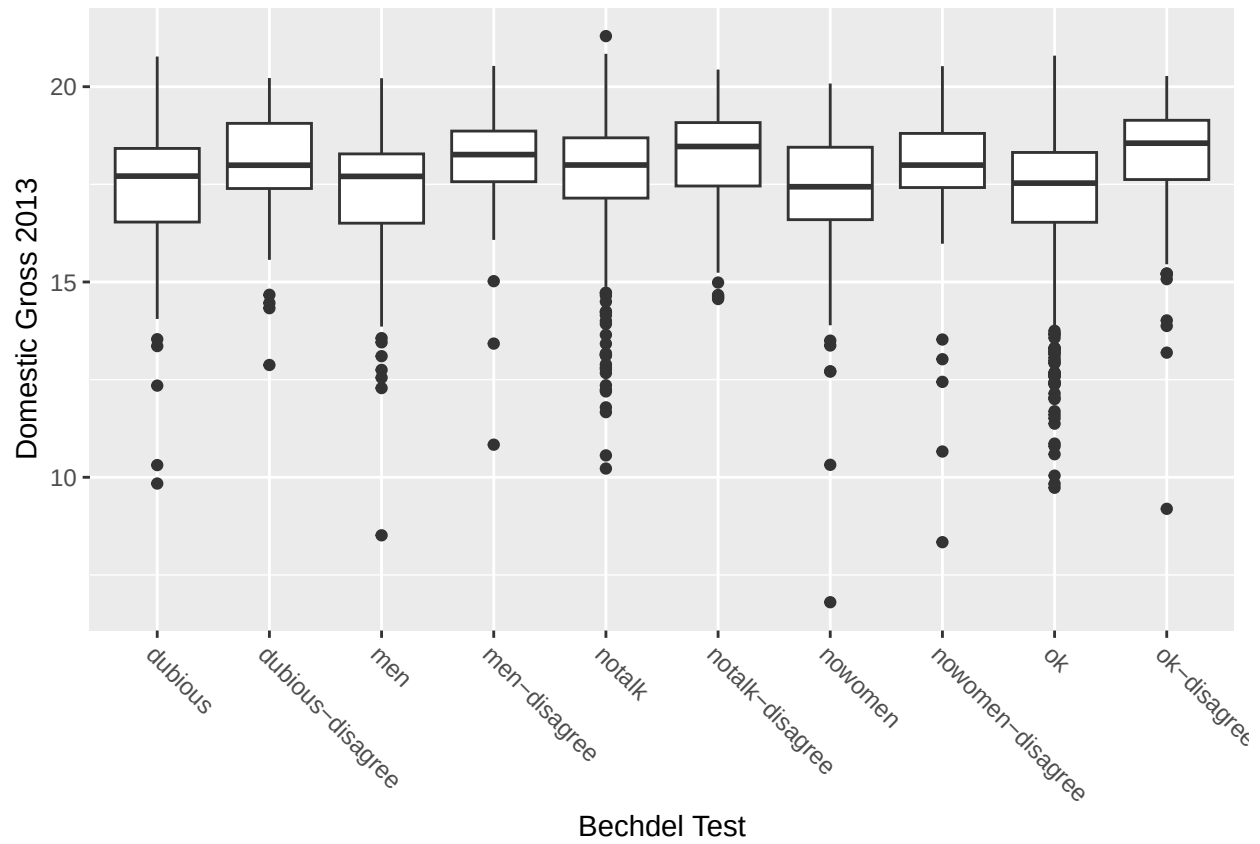


```
# using ggplot
ggplot(bech, aes(binary, domgross_2013)) + geom_boxplot() + labs(y = "Domestic Gross 2013",
  x = "Bechdel Test") #creates boxplots based on values of binary
```

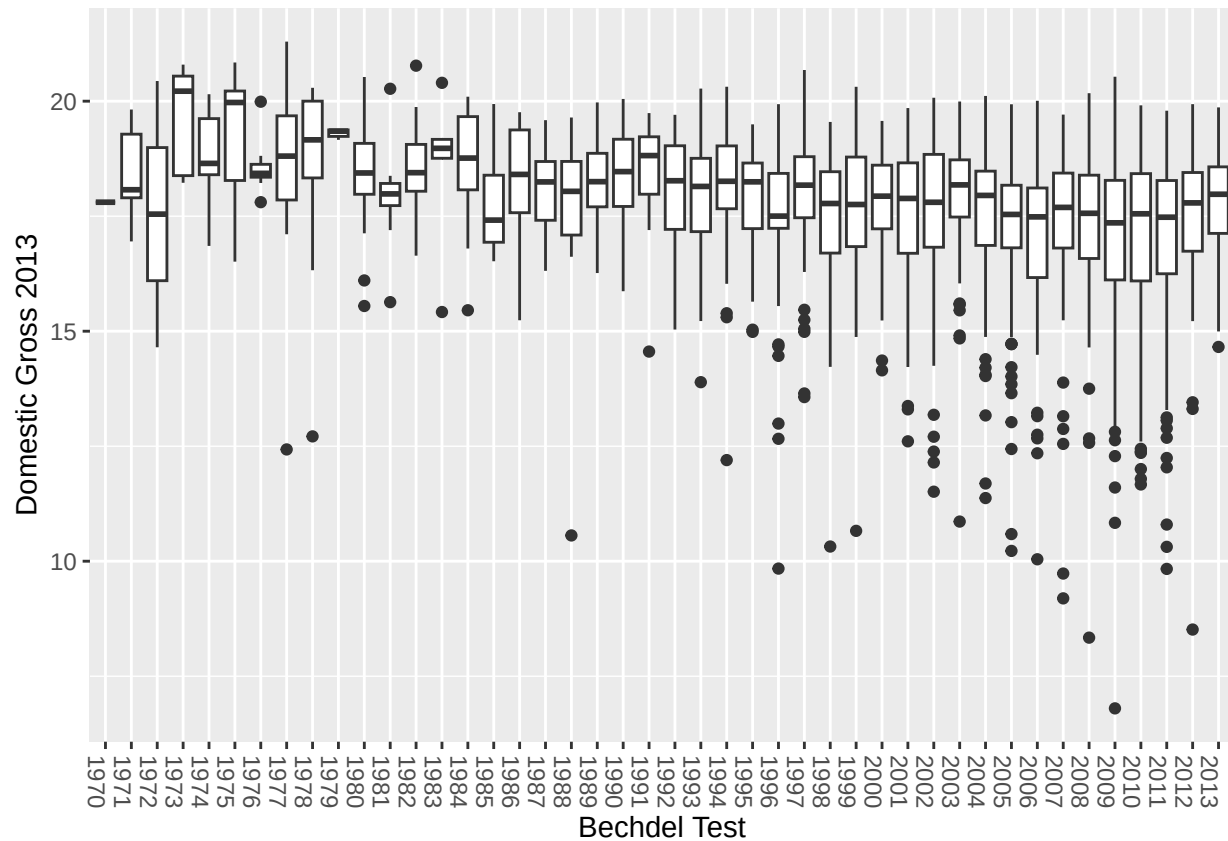


```
# to illustrate we can use this for variety of categories (and logged)
ggplot(bech, aes(test, logdomgross13)) + geom_boxplot() + labs(y = "Domestic Gross 2013",
```

```
x = "Bechdel Test" + theme(axis.text.x = element_text(angle = -45, hjust = 0))
```



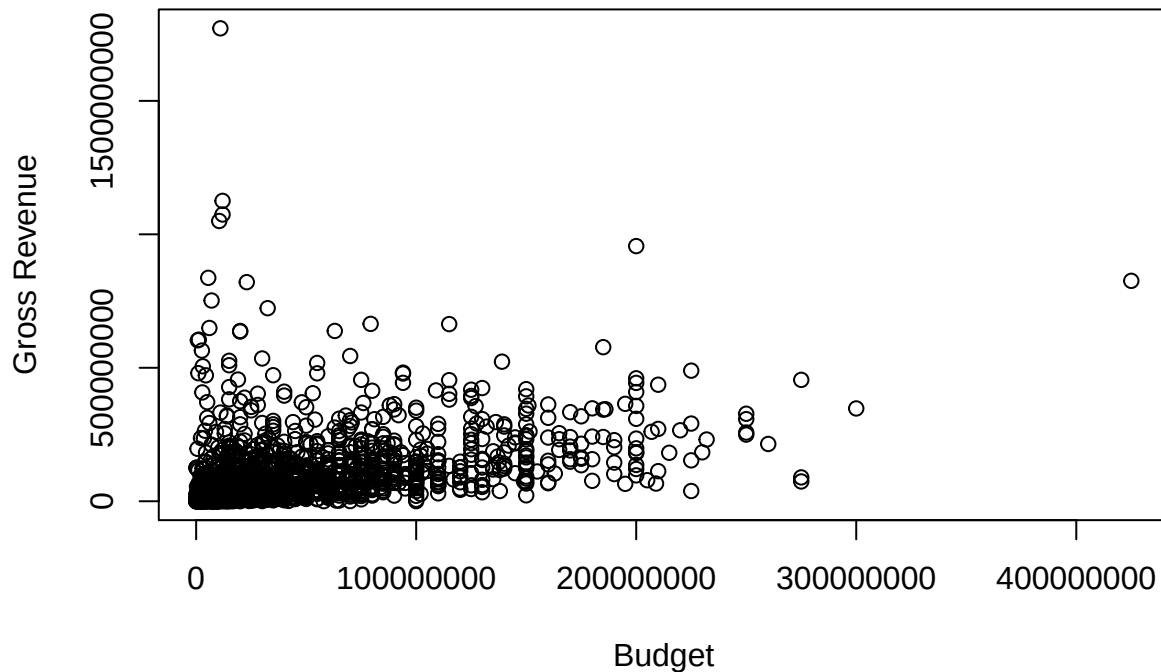
```
ggplot(bech, aes(as.factor(year), logdomgross13)) + geom_boxplot() + labs(y = "Domestic Gross 2013",  
x = "Bechdel Test") + theme(axis.text.x = element_text(angle = -90, hjust = 0))
```



FINDING RELATIONSHIPS *(Let's see if bigger budgets are related to a higher gross)*

```
par(mfrow = c(1, 1)) #show one graph at a time
plot(bech$budget, bech$domgross_2013, main = "Gross Revenue and Budget", xlab = "Budget",
     ylab = "Gross Revenue")
```

Gross Revenue and Budget



How would we create the same plot using ggplot? (hint: look at last week's code)

IT'S HARD TO SEE A PATTERN FROM THE PLOT, SO LET'S MAKE A LINEAR REGRESSION MODEL: REMEMBER THE MODEL IS $\text{lm}(Y \sim X, \text{data}=\text{nameofdata})$

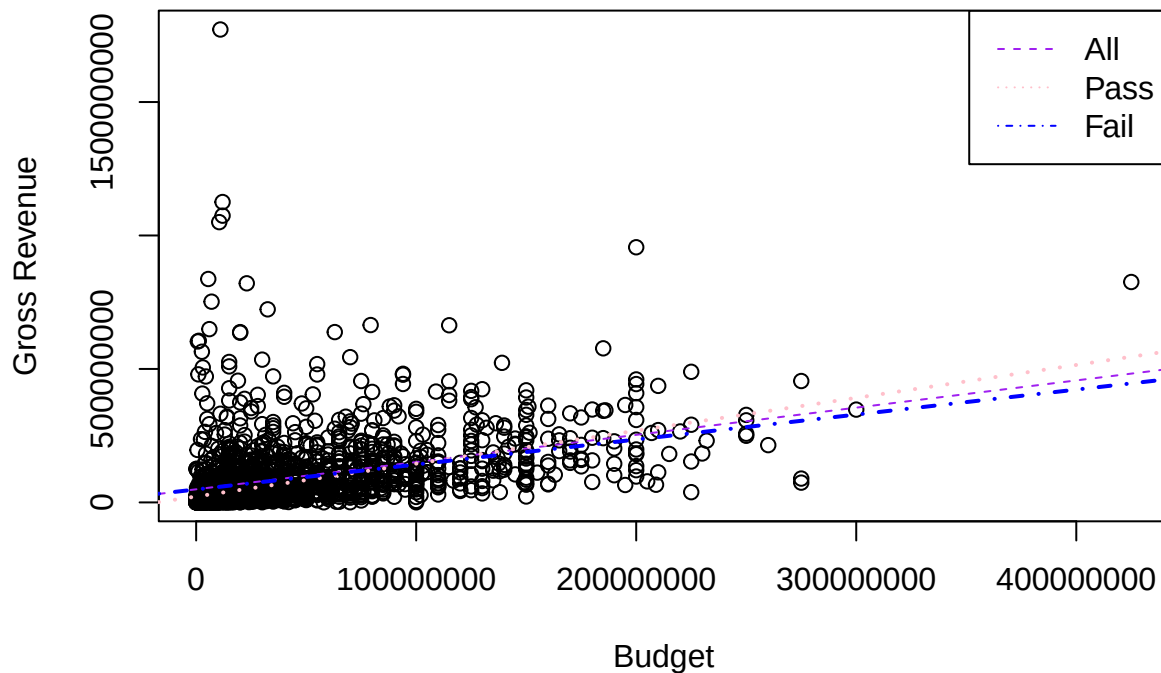
```
plot(bech$budget, bech$domgross_2013, main = "Gross Revenue and Budget", xlab = "Budget",
     ylab = "Gross Revenue")
model1 <- lm(domgross_2013 ~ budget, data = bech) #Making an object for a linear regression model. Not
summary(model1)
```

```
##
## Call:
## lm(formula = domgross_2013 ~ budget, data = bech)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -255385370 -53329248 -32315156  14241205 1711353507
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 49113273.320  3768631.907   13.03  <2e-16 ***
## budget         1.020      0.057   17.89  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1160000000 on 1774 degrees of freedom
## Multiple R-squared:  0.1528, Adjusted R-squared:  0.1524
## F-statistic: 320 on 1 and 1774 DF, p-value: < 2.2e-16
```

```
abline(model1, col = "purple", lty = 2) #This puts the trend line
```

```
abline(lm(pass$domgross_2013 ~ pass$budget_2013), lty = 3, lwd = 2, col = "pink") #for fun add lines f
abline(lm(fail$domgross_2013 ~ fail$budget_2013), lty = 4, lwd = 2, col = "blue")
legend("topright", legend = c("All", "Pass", "Fail"), lty = c(2, 3, 4), col = c("purple",
"pink", "blue"))
```

Gross Revenue and Budget



What do the different slopes mean? What does this tell us about the profitability of including in the movie industry?

Another Subset: We may also want to exclude outliers

```
middle <- subset(bech, budget_2013 >= 1000000 & budget_2013 < 10000000) #only budgets between 1 and 10
summary(middle$budget_2013)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 1014639 2827594 5178454 5217895 7600121 9897188
```

```
# do the relationships change when we control for outliers(hint: use copy and
# paste plus find and replace)
```

```
# What about when we use the log of gross
```